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Techno-economic assessment and optimization framework with energy storage for hybrid energy resources in base transceiver stations-based infrastructure across various climatic regions at a country scale

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ABSTRACT

Renewable energy presents a sustainable solution for tackling both energy access and environmental issues. Hybrid off-grid systems appear to be a promising concept for addressing energy security and balancing issues in many localities across the world in the presence of unreliable supply from the utility grid. Therefore, the undermining of enormous renewable potential in developing countries especially Pakistan compelled to rely on traditional energy sources, which give rise to notable socio-economic challenges for many sectors. In the context of the telecom sector especially Base Transceiver Stations (BTS), hybrid renewable energy systems can ensure a stable power output by combining different energy sources, which is crucial for providing a reliable and uninterrupted power supply. Furthermore, the deployment of such a system has the potential to lower the operational expenses and environmental concerns of BTS that presently heavily depend on diesel generators with battery storage backup. This study introduces a comprehensive framework for implementing a large-scale hybrid (solar, wind, and battery) based standalone systems for the BTS encapsulation telecom sector. Based on the actual load profiles, the framework presents a comprehensive techno-economic evaluation of 35 independent sites located in the North, South, and Central regions of Pakistan with the aims of minimize the primary parameters including net present cost (NPC), Levelized cost of energy (LCOE), operational costs, diesel generator' operating hours, and capacity shortage fraction. Optimal results demonstrate that the photovoltaic system coupled with a diesel generator having the battery storage system remains the most feasible option for Central and Northern regions with an average LCOE of 0.1267 $/kWh to 0.1815 $/kWh. Due to the significance of wind profile, mainly in the South region, a hybrid energy system (solar, wind, diesel, and battery) remains the most optimal option with an average LCOE of 0.1096 $/kWh to 0.1294 $/kWh.

The proposed study can serve as a guide for decision-making and policy optimization for the Sustainable Development Goals (SDGs) as well as for public and private investors.

1. Introduction

In today's industrialized society and modern civilization, energy is considered one of the most fundamental and expanding human necessities, operating as the main engine powering the global economy. The observation of ideal generating patterns comes from combining renewable energy sources (RES) and suitable storage. According to the International Energy Agency (IEA), Global energy consumption was

Abbreviations: BTS, Base Transceiver Stations; DGH, DG hours; BCR, battery cycle rate; HOMER, hybrid optimization model for electric renewable; LCE, life cycle emission; HES, hybrid energy system; DG, diesel generator; PV, photovoltaic; RE, Renewable Energy; EE, excess energy (kWh); FC, fuel cell; BA, battery; PBP, payback period; FC, fuel consumption; NPC, net present cost; TLP, total loan payment; LF, load following; CC, charging cycle; LCOE, levelized cost of energy; IRR, internal rate of return; CF, capacity factor; HRER, hybrid renewable energy resource; BESS, battery energy storage system; COE, cost of energy; HES, hybrid energy systems; DER, distributed energy resources; COE, cost of energy ($); WT, wind turbine; DPSP, deficiency of power supply probability; WE, wind energy; MG, microgrid; TLC, thermal load controller; HRES, hybrid renewable energy system; COUS, cost of unit saved; SEM, sustainable energy mix; COER, cost of energy replaced; BA, bill amount; GF, generator fuel; FR, fuel rate; TV, tax value/cost of tax; RF, renewable fraction; BOGF, bill of generator fuel; ROI, return on investment; GHG, greenhouse-gases.

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anticipated to increase by around 4 % in 2021 and continues to increase, returning to pre-pandemic levels as economic activity improves. Furthermore, meeting this energy increasing demand, especially with non-sustainable resources like fossil fuels, presents various challenges. Nearly 80 % of the world's foremost energy comes from fossil fuels. Between 2015 and 2040, global energy consumption is expected to increase at a rate of about 2.3 % annually, posing a threat to the atmosphere's CO_2 concentration. Since the fourth industrial revolution has begun, the atmospheric CO_2 equivalent concentration has roughly doubled [1].

Various businesses particularly mobile communication are functioning to serve many customers due to technology development To meet the growing demand for telecommunications services, cellular network operators experience from a significant serious challenges due to the absence of reliable grid electricity in rural or isolated areas. Therefore, telecom equipment runs on diesel generators (DG) or batteries during peak demand periods, increasing the risk of global warming and CO_2 emissions. This emphasizes using RES, like solar, wind, and tidal power, to power telecom towers [2]. The largest source of plant-level energy expenses for off-grid tower sites is the cost of diesel fuel for generators, which makes up over 80 % of all energy expenditures. To achieve the best return on investment, operators and vendors must closely examine and assess design criteria, maintenance, and operations scenarios, the energy efficiency of BTS, greenhouse gas emissions, and technology evolution capabilities [3].

Despite the rampant population's increasing acceptance of mobile telephony, the base transceiver station power supply may make it more difficult to reap the benefits of its services. Following this, the mobile telecommunications operators install diesel generators as supplemental energy sources to reduce unstable power supply but at the same time experience serious financial, logistical, and environmental concerns [4]. Hybrid energy systems based on standalone or grid-connected energy provisioning for the telecom sector can play a vital role in overcoming the aforementioned challenges. However, as there are variations in a renewable generation because sun radiation and wind speed are unpredictable [5], contrary to on-grid systems, standalone solar and/or wind systems are not able to meet the entire 24-h need for electricity. As a result, there must be HRES that combine HES with storage technologies which improves and increases the reliability of the system. Hence, the most ideal alternative for lowering the use of diesel as well as for resolving power shortages is a hybrid system that combines solar, wind (if available), and DG with energy storage devices (battery) [6].

Pakistan has rich potential of different type of RESs especially solar and wind in southern and southwest which can be utilized to power different BTS in furlong areas. This article examined various cellular BTS powering options. According to a techno-economic analysis [7], an alternative for cities is a hybrid energy system that uses solar PV, small-scale wind, diesel, and batteries. In addition, it produces the least amount of CO_2 and is thus pollution-free. When solar and wind power systems are combined on a telecom site, the electrical energy produced by the PV-DG and wind systems is directly fed to the base transceiver station load with a battery storage system and charge controller. The essential duty of a battery is to store energy that can be utilized when the production of sunlight is minimal at night or when only wind energy is operating. Because the reliability of solar and wind energy is 100 %, a DG generator must be utilized as a backup so that the diesel generator may supply the base load in an emergency.

In Pakistan, all BTS sites depend purely on diesel generators with battery storage systems. However, in this paper, 35 different BTS sites are discussed with the integration of Solar, wind, and diesel generator with the battery storage system. Due to the high diesel rate, BTS sites are usually unable to meet their baseload economically as well as environmentally friendly due to increased carbon emissions. Thus, there is a dire need to analyze the techno-economic feasibility of hybrid energy systems (solar, wind, diesel generator, and battery) to replace the uneconomical, unreliable and non-sustainable diesel generator. This techno-economic assessment of HRES has been quantified using various assessment indicators, including net present cost (NPC), operating cost, internal rate of return (IRR), return on investment (ROI), Levelized cost of energy (LCOE), payback period (PBP) and GHG emissions. A substantial amount of research on various HRESs has been performed to create the ideal Sustainable Energy Mix.

2. Literature review

Several but limited studies have been presented in the literature to discuss the techno-economic evaluation of hybrid energy systems (HES) for electrification of different localities and load centers. Researchers in ref. [8] focused on the techno-economic assessment of HES optimization for distributed power generation in metropolitan settings. The results indicate sufficient space for implementing hybrid energy systems in urban areas to supply additional electricity during blackouts. The suggested system's techno-economic viability was thoroughly examined and evaluated throughout the research using a hybrid optimization model for electric renewables (HOMER) toolbox. The author [9] used data from solar radiation, temperature difference, and wind speed to address an improvement in the techno-economic ideal size of an off-grid MG system in the presence of autonomous days. This approach to evaluating system reliability is built on the DPSP. The EC and the TNPC are used to investigate the system cost. To illustrate how autonomous days affect system performance and price, we looked at two MG system configurations: (PV, battery) and (PV, wind, battery). The diesel generator has been employed as a backup power source because of its efficiency in satisfying load demand and low-cost relative to battery capacity. Comparison research has indicated that a large battery capacity can increase the MG system's initial investment. Rapid changes and analyses have been carried out worldwide to encourage the electrification of rural communities. These energy sources include the sun, wind, and small amounts of hydropower.

Pakistan has a substantial and assorted renewable energy potential. Therefore, Pakistan's northern and western regions have significant potential for photovoltaic, wind, hydel, and biomass energy. Like the Northeast zone, the Southwest zone has an excellent solar profile and is a great place to construct residential and commercial-scale PV systems. The literature on techno-economic analysis to address the power discrepancy has some significant studies. The authors of [10] create a rigorous framework for planning a rural energy system that utilizes readily available energy supplies to meet the electrical requirements of a particular site. The proposed framework is implemented in Dera Ismail Khan, a remote Pakistani community. This study explored several hybrid system configurations, including PV/DG/batteries and time-constrained grid availability, to satisfy the region's electrical power requirements. The researcher [11] evaluated a photovoltaic with biomass energy system technologically and economically to support the agricultural and electrical needs of a lesser community in the district, Layyah of Punjab province. Similarly, article [12] evaluates the techno-economic viability and sustainability of various PV-WE-BA based hybrid system configurations for fourteen sites in Gilgit Baltistan (GB). Additionally, the research assessed the forest's ability to absorb GHG emissions.

In this work [13], a grid-free HREM was developed to satisfy the electrical requirements of a remote hamlet in South India. The analysis was finished using the HOMER software program, and an economic HREM setup was made. Based on the parameters of the load profile and the RERs that were available at the study site, simulations were run. According to simulated analyses that demonstrate that component sizes, cash flow summaries, electrical energy generation, and greenhouse gas emissions were considered, the HREM system composed of PV/DG/BESS was found to be the most practicable.

From the literature review, it is clear that no research is done on the techno-economic assessment of renewable integration with BTS sites in Pakistan. To the best of the authors' knowledge, earlier research looked at whether the suggested system optimally relied on just two or three

factors in the development, primarily LCOE, and NPC. Prior research concentrated on locations with comparable topographical and climatic circumstances. Since no study has been available that emphasizes the varied nature of sites at a preliminary national level, previous research has been limited to several areas. Table 1 thoroughly compares the contributions/novelty of proposed investigations compared to previously conducted studies. In earlier published research, the choice of sites has been chiefly based on a random analysis, with the same energy sources employed for study at each location. Similar to how most research chooses standard and common components for each site or uses the hit-or-trial strategy to get close to ideal outcomes, none of which are available on the commercial market. On the other hand, storage technology applications vary according to the climate zone of each under-researched site. It is important to note that most previous studies assess the suggested analysis under ideal circumstances. For instance, the default discount value and inflation rates are considered in most research. While various losses related to components like photovoltaic panels, batteries, and wind turbines are disregarded during analysis, systems are modeled with a zero-loss factor. In real usage, a variety of losses have an impact on how well wind turbines operate.

The suggested study's innovative component includes the simultaneous consideration of several decision characteristics to find an ideal system. Therefore, decision-making criteria include LCOE, NPC, ROI, PBP, reliability or capacity shortage percentage (CSF), and unmet demand (UMD). The use of actual input datasets is another innovative feature of this work. For analysis, inflation and discount rates used for the fiscal year 2021–2022, which reflect the actual state of the nation, are employed. The suggested study concentrated on locations with various topographical and climatic features. The losses are computed using commercially installed local wind farms. In summary, the proposed research is innovative since it considers all the corrected flaws identified in the literature.

2.1. Motivation of the study

Interest in HRES has increased due to the fast-rising demand for power, environmental concerns, the diminishing availability of fossil fuels, unpredictable prices, and the need to supply electricity to off-grid

Table 1
Recent studies on techno-economic analysis of hybrid energy systems and contribution of proposed study.

Sr. no.	Location	Publication year	BTS sites	Method	Technical characteristics components					Objective functions and explicitly considered variables				Load type
					PV	WE	DG	BC	BA	LCOE	IRR	ROI	SA	
1	Australia [14]	2022	✘	HOMER	✔	✔	✔	✘	✔	✔	✔	✘	✔	IL
2	Buea Cameroon [15]	2022	✘	HOMER	✔	✔	✔	✘	✘	✔	✔	✘	✘	DOM
3	Gorgan city, Iran [16]	2022	✘	IGOA	✔	✔	✘	✘	✔	✔	✘	✘	✔	COM
4	India [17]	2022	✘	HOMER	✔	✔	✔	✘	✔	✔	✔	✘	✔	DOM
5	Nigeri [18]	2022	✘	HOMER	✔	✔	✘	✘	✔	✔	✘	✘	✔	DOM
6	Gujrat, India [19]	2021	✘	HOMER	✔	✔	✘	✘	✔	✔	✘	✘	✔	AGR
7	Northern India [20]	2021	✘	HOMER	✔	✔	✔	✘	✔	✔	✔	✘	✔	RSD
8	Pakistan [10]	2021	✘	HOMER	✔	✔	✘	✘	✔	✔	✘	✘	✔	COM
9	Rajshahi, Bangladesh [21]	2021	✘	HOMER	✔	✔	✔	✘	✔	✔	✔	✘	✔	RSD
10	Delhi, India [22]	2021	✘	HOMER	✔	✘	✘	✘	✔	✘	✘	✘	✔	RSD
11	Sousse, Tunisia [9]	2021	✘	HOMER	✔	✔	✘	✘	✔	✔	✘	✘	✔	COM
12	Saudi Arabia [23]	2020	✘	HS,PSO	✔	✔	✔	✘	✔	✔	✔	✘	✔	DOM
13	Egypt [24]	2022	✘	HOMER	✔	✔	✘	✘	✘	✔	✘	✘	✘	COM
14	Karnataka, India [25]	2020	✘	HOMER	✔	✘	✘	✘	✔	✘	✘	✘	✔	COM
15	Lanzhou, Gansu, China [26]	2020	✘	HOMER	✘	✔	✔	✔	✔	✔	✔	✔	✔	IND
16	Koh Jik island, Thailand [27]	2020	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	IL
17	Baghdad, Iraq [28]	2020	✘	HOMER	✔	✘	✘	✔	✔	✘	✘	✔	✔	RSD
18	Europe [29]	2020	✘	GA,PSO	✔	✔	✘	✘	✔	✔	✘	✘	✔	DOM
19	Benin, Africa [30]	2020	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	DOM
20	Uttar Pradesh, India [31]	2019	✘	HOMER	✔	✘	✘	✔	✔	✘	✘	✔	✔	DOM
21	Chungbuk, South Korea [32]	2019	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	COM
22	Jubail, Saudi Arabia [33]	2019	✘	HOMER	✔	✔	✔	✔	✔	✔	✔	✔	✔	IND
23	Jiuduansha, China [34]	2019	✘	HOMER	✔	✔	✘	✔	✔	✔	✘	✔	✔	IL
24	Madhya Pradesh, India [35]	2018	✘	GA,PSO	✔	✔	✔	✘	✔	✔	✔	✘	✔	AGR
25	Kallar Kahar, Pakistan [36]	2018	✘	HOMER	✔	✔	✘	✔	✔	✔	✘	✔	✔	DOM
26	Izmir Province, Turkey [37]	2018	✘	HOMER	✔	✔	✔	✔	✔	✔	✔	✔	✔	RSD
27	Saudi Arabia [38]	2018	✘	HOMER	✔	✔	✘	✔	✘	✔	✘	✔	✘	COM
28	Godagari, Bangladesh [39]	2018	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	DOM
29	Himalayas, Northeast India [40]	2018	✘	HOMER	✔	✔	✔	✔	✔	✔	✔	✔	✔	DOM
30	South Korea [41]	2017	✘	HOMER	✔	✔	✔	✔	✔	✔	✔	✔	✔	COM
31	Kilis, Southern Turkey [42]	2017	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	RSD
32	Saudi Arabia [43]	2017	✘	HOMER	✔	✔	✘	✔	✔	✔	✘	✔	✔	DOM
33	Layyah, Punjab, Pakistan [11]	2017	✘	HOMER	✔	✘	✘	✔	✔	✘	✘	✔	✔	AGR
34	Varanasi, India [44]	2017	✘	HOMER	✔	✘	✘	✘	✘	✘	✘	✘	✘	DOM
35	Tsumkwe, Namibia [45]	2017	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	RSD
36	Sarawak, East Malaysia [46]	2017	✘	HOMER	✔	✘	✘	✔	✔	✘	✘	✔	✔	DOM
37	Colombia [47]	2017	✘	HOMER	✔	✔	✔	✔	✔	✔	✔	✔	✔	DOM
38	Shafar, Yamen [48]	2017	✘	HOMER	✔	✔	✔	✔	✘	✔	✔	✔	✘	DOM
39	Sabah, Malaysia [49]	2017	✘	HOMER	✔	✘	✔	✔	✔	✘	✔	✔	✔	DOM
40	Hendijan County, Iran [50]	2017	✘	HOMER	✔	✔	✔	✔	✔	✔	✔	✔	✔	IND
41	Chile [51]	2017	✘	HOMER	✘	✔	✘	✔	✘	✔	✘	✔	✘	RSD
42	Saudi Arabia [52]	2017	✘	PSO	✔	✔	✘	✘	✔	✔	✘	✘	✔	DOM
43	Australia [53]	2017	✘	PSO	✔	✔	✘	✘	✘	✔	✘	✘	✘	DOM
44	**Pakistan, Proposed Study**	**2022**	✔	**HOMER**	✔	✔	✔	✔	✔	✔	✔	✔	✔	**TELEC**

Note: PV: Photo-voltaic, WE: Wind energy, DG: Diesel Generator, DC: Bi-directional converter, BA: Battery, LCOE: Levelized cost of energy, IRR: Internal rate of return, SA: Sensitivity analysis, IL: Island load, DOM: Domestic, COM: Commercial, AGR: Agricultural, RSD: Residential, IND: Industrial, TELEC: Telecom BTS load.
The bold format is used to show the contribution of the proposed study. However, in session report, the ✔ and ✘ are not shown properly.

systems. However, a comprehensively sustainable HRES design must consider technological, economic, environmental, and social concerns. Additionally, a comparison of several dispatch techniques in terms of technical, financial, ecological, and social considerations is necessary to establish an effective HRES dispatch strategy.

This study is performed not just to lower stakeholder costs associated with meeting load demand but also to improve social standing while reducing environmental pollution. This research inspired the authors to research and publish a study on the integrated techno-economic-environmental-socio-technical design of the hybrid renewable energy system with a suitable dispatch strategy for household and telecommunication loads throughout Pakistan.

2.2. Major contributions

The paper aims to provide a techno-economic feasibility analysis of batter-storage based hybrid renewable energy sources-based infrastructure to feed the telecom sector load in 34 independent sites of Pakistan. The major contributions of this proposed work are summarized as follows:

i. On a country's scale, no study has been reported to discuss the integration of renewable energy sources to power BTS. This study contributes to integrating RES to power BTS based on real-time load profiles.
ii. This study provides a policy roadmap to reduce the BTS associated risks with the aim of minimizing economic and environmental challenges.
iii. Optimize the RES to fulfill the energy demand of BTS to increase the reliability and service restoration of BTS systems.
iv. During to interruptible power supply from grid and expensive fuel prices to run the diesel generator, current BTS suffers from the reliable provision of power which consequently suffers the end customers. This may affect the socio-economic conditions of the customer. To tackle these conditions, this paper contributes to recover the reliability of the system by provision of renewable based hybrid energy systems for telecom sector.
v. Component prototypes and storage technology are employed based on precursory screening to produce superior results. Numerous losses related to wind turbines are also considered during optimization to depict a realistic viewpoint.

This paper is organized into 7 sections; the introduction and literature review with significant contributions are in Section 1 and Section 2, while Section 3 includes research methodology, and Section 4 presents the system modeling followed by the results obtained along with relevant discussions presented in Section 5. Section 6 provides policy recommendations and implications with concludes the study with future work in Section 7.

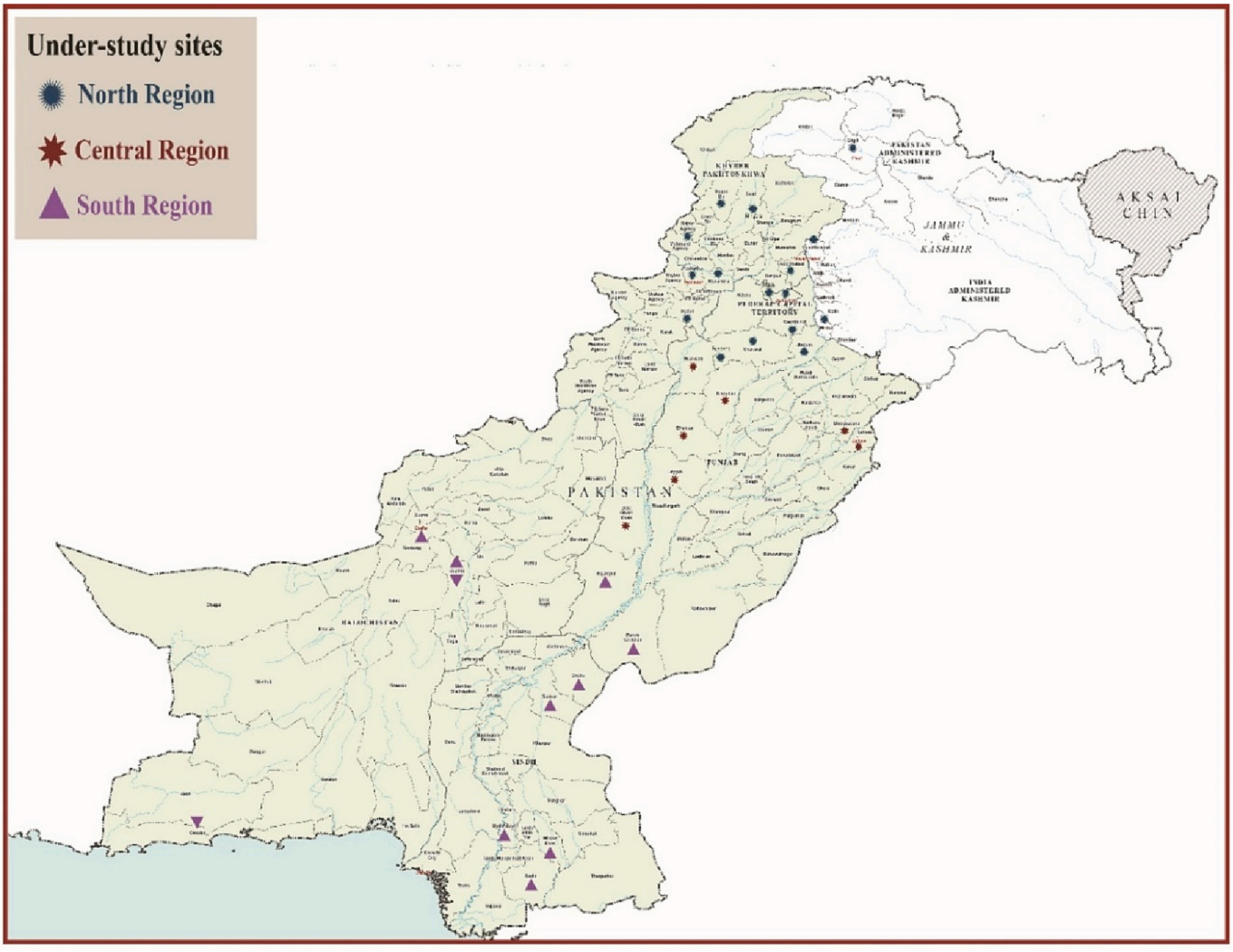


Fig. 1. Geographical illustration of selected BTS sites throughout Pakistan.

3. Research methodology

This investigation is carried out by using BTS sites around Pakistan. The complete graphical map presentation of nominated sites in Pakistan is presented in Fig. 1. The whole region of Pakistan is divided into three zones i.e., the North region, the Central region, and the South region depending upon the climatic conditions with the geographic and socio-economic parameters. The current inflation and discount rates for the year 2021–2022 were used, which imitate an accurate stance of the country, i.e., high inflation and fluctuating policy rates according to the State Bank of Pakistan (SBP), are chosen in this research. The primary input data sets, presumptions, and optimization parameters with entire sources are summarized in Table 2. In this study, RESs-based BA-WE-PV hybrid MG systems are examined based on the probable resource availability for each site. A diesel generator (DG) is utilized as a backup energy source due to its sustainability and 24-hour dependability.

The planned study is performed in HOMER Pro based on evaluating its sustainability and techno-economic impact. In terms of intricate settings, reliability, and hourly time step simulation, HOMER Pro has many benefits. Environmental assessment is done based on GHG emissions for sustainability analysis objectives. The Energy Sector Management Assistance Program (ESMAP) and World Bank execute the primary broadcast for renewable resource potential, i.e., solar irradiations and wind profile estimation across the whole region, using various tools like the world solar and global wind atlas. NASA's database was used to confirm the existence of sites with good resource potential. In this study, 34 BTS sites were initially shortlisted from over the Pakistan region, as shown in Fig. 1.

Of these 34 BTS sites, a total of seven sites are taken from the central region of Pakistan named Lahore, Sheikhupura, Bhakkar, Khushab, Mianwali, DG Khan, and Layyah. From the north region, there are sixteen sites selected named as Chakwal, Islamabad, Jehlum, Rawalpindi, Talagang, Taxila, Bajaur, Dir, Kohat, Nowshera, Peshawar, Abbottabad, Gilgit, Mingora, Mirpur, Muzaffarabad. From the south region of Pakistan, a total of eleven sites are selected for this study named as Karachi, Badin, Hyderabad, Mirpur Khas, Ghotki, Rajan Pur, Rahim Yar Khan, Sukkur, Gawadar, and Quetta. The detailed research methodology is discussed in Fig. 2.

Table 2
Desired inputs, datasets, assumptions, and other crucial variables were considered in this study.

Sr. no.	Inputs	Value	Source
1	Electrical load		
	Scaled annual average	Site-specific	Pakistan Telco
	Timestep & day-to-day variability	0 %–1 %	Not applicable
	Load profile	Commercial	Built-in HOMER
	Peak months	Constant load	Not relevant
2	Resource parameters		
	Temperature	NASA database	Coordinates dependent
	Solar irradiance	NASA database	Coordinates dependent
	Wind speed	NASA database	Coordinates dependent
	Wind turbine losses	LF: 21.02702 %	Source: [54]
3	System economics		
	Discount rate	9.75 %	State Bank of Pakistan [54]
	Inflation rate	8.50 %	State Bank of Pakistan [54]
	Annual capacity shortage fraction	0.00 %	Not applicable
	Project life	25 yrs.	Not applicable
	Minimum renewable fraction	0.00 %	Not applicable
4	Fuel and System costs		
	Diesel fuel price	$1.03/L	Market value: Oct 05,2022
	Cost parameters	On-ground data ($/kW)	Previous studies
5	Operating reserve		
	Load in current time step	10.00 %	Not applicable
	Solar power output	80.00 %	Not applicable
	Wind power output	50.00 %	Not applicable
6	Optimization settings		
	Optimization timestep	60 min	Not applicable
	Focus factor	50	Not applicable
	System design precision	0.0100	Not applicable
	NPC precision	0.0100	Not applicable
	Battery Autonomy	Less than 2 h	Not applicable
	Maximum renewable penetration	Greater than 55 %	Not applicable
	Maximum simulation per optimization	10,000	Not applicable

3.1. Objective functions and performance indicators

Following that, the proposed study is performed in HOMER Pro; both the original grid search algorithm (OGSA) and the proprietary derivative-free algorithm (PDFA) are used in this research and rely upon objective functions like Net present cost (NPC) and Levelized cost of energy (LCOE) during optimization and designing, though also enhancing the injection of renewable diversity. The cost of a system's installation and operation throughout a project is commonly referred to as NPC or LCC [12]. The entire cash inflow throughout the chosen term "t," the starting investment, and the discount rate "r" all determine NPC [1]. The energy cost per production unit across a product's lifetime is known as LCOE [23]. In Eqs. (1), (4), and (7), the objective functions (O. Fs) are mathematically explored.

$$\mathrm{OF}-1\ (\mathrm{Minimum})::NPC_T=\sum_{t=1}^{T}C_t\left((1+r)^{-t}-\frac{C_o}{C_1}\right) \tag{1}$$

Eq. (1) illustrates,

$$min\{x.NPC_{PV}+y.NPC_{WT}+z.NPC_B\} \tag{2}$$

$$\mathrm{LCC}=\sum\left(\mathrm{C_{cap}}+\mathrm{C_{rep}}+\mathrm{C_{fuel}}+\mathrm{C_{0\&M}}\right)-\mathrm{C_{salvage}}$$

$$C_{salvage}=C_{rep\times}\ (R_T/T) \tag{3}$$

In Eq. (1), r is known for a discount rate that is followed by the desired circumstances of the country, T is the number of years, C_o is salvage cost and C_t is called total annualized cost. In Eq. (3), R_T denotes the remaining life of the project with C_{rep} as replacement cost and T as the lifetime of the project.

$$\mathrm{OF}-2\,(\mathrm{Minimum})::LCOE_T\left(\frac{\$}{KWh}\right)=\sum_{t=1}^{T}\frac{\frac{1}{(1+r)^T}\left(C_t-C_o(1+r)^t\right)}{E_P\times(1+r)^{-t}}\times C_{RF}(r,T) \tag{4}$$

$$C_{RF}=\frac{r\times T}{rT-\left(\frac{1}{1+r}\right)} \tag{5}$$

Eq. (3) illustrates,

$$min\{x.LCOE_{PV}+y.LCOE_{WT}+z.LCOE_B\} \tag{6}$$

$$OF-3\ (Minimum)::R.Div_{inj}\ \{PV,WT,B\} \tag{7}$$

In Eq. (4), load for all individual sites is denoted by *Ep*. C_{RF} is called capital recovery factor [55] and this factor is used to find the coefficients *x, y,* and *z*, which used in Eq. (3) and Eq. (6). The optimal solution minimizes the LCOE and NPC in this research study. Therefore, these are also financial parameters closely linked with performance. The time (years) required to cover the project's investment cost is called the payback period, denoted by PBP. The formula used to calculate the P_{BP} is given in Eq. (8).

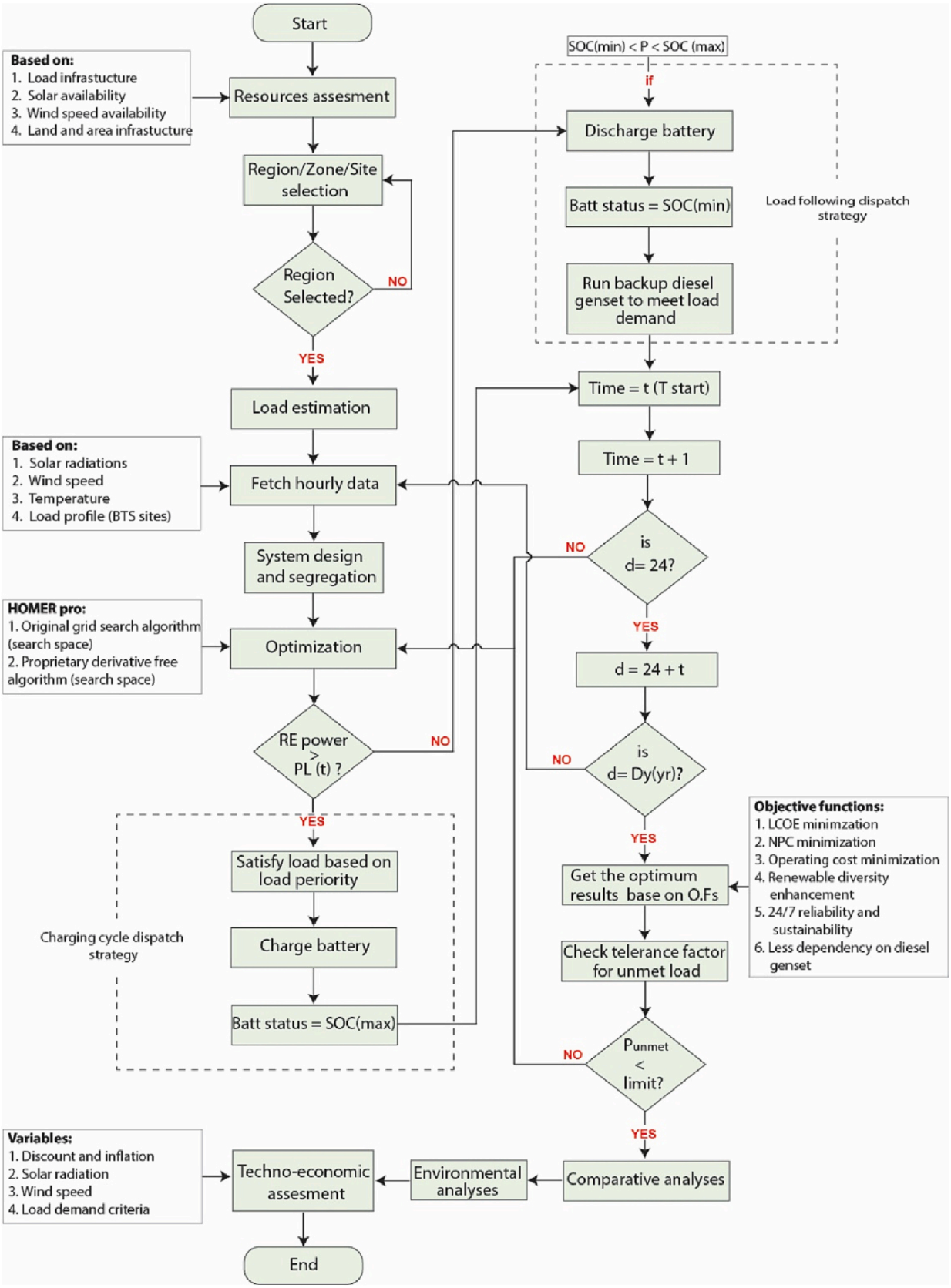


Fig. 2. A flowchart of the research methodology and the function of the dispatch schemes.

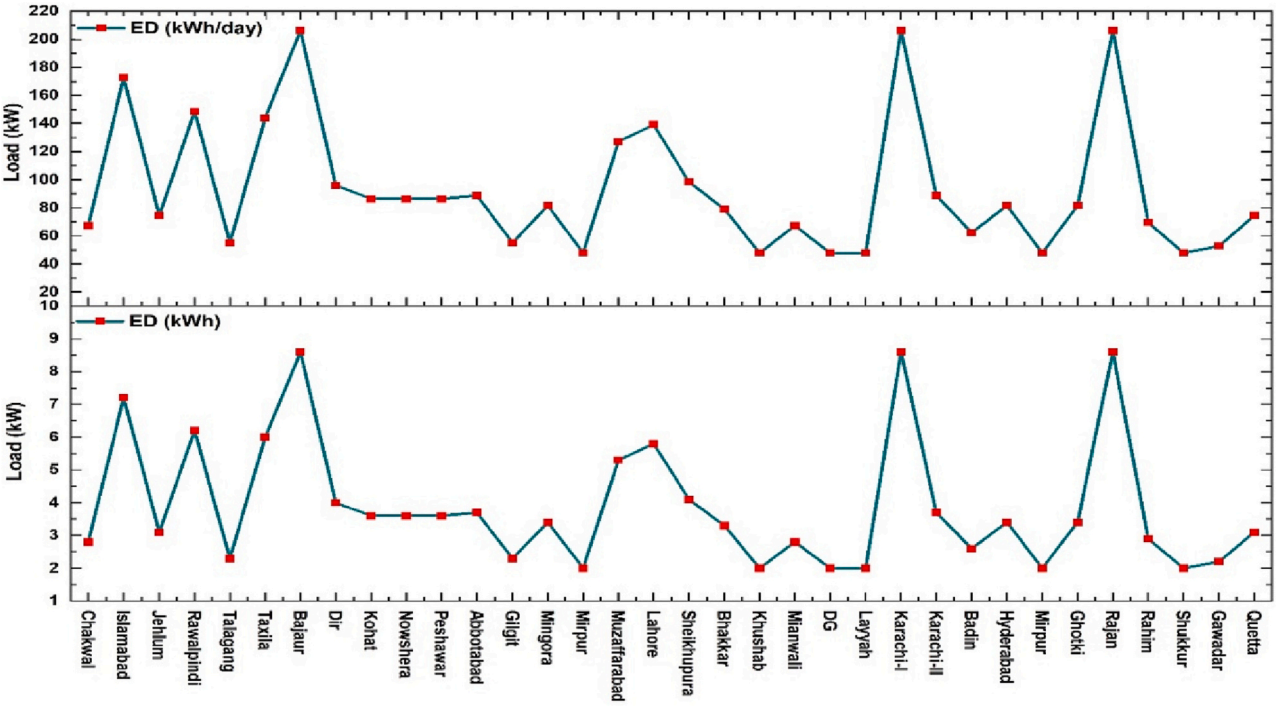


Fig. 3. Hourly and daily Load profile of all BTS sites.

Table 3
Pakistan: relative information for selected BTS sites with their energy demands.

Sr. no.	Zone / region	Site name / site designation	Coordinates	BTS load demand	
				E_D (kWh)	E_D (kWh/day)
01	North	Chakwal	32.9328° N, 72.8630° E	2.8	67.2
02	North	Islamabad	33.6844° N, 73.0479° E	7.2	172.8
03	North	Jehlum	32.9425° N, 73.7257° E	3.1	74.4
04	North	Rawalpindi	33.5651° N, 73.0169° E	6.2	148.8
05	North	Talagang	32.9172° N, 72.4081° E	2.3	55.2
06	North	Taxila	33.7463° N, 72.8397° E	6.0	144
07	North	Bajaur	34.7865° N, 71.5249° E	8.6	206.4
08	North	Dir	35.1977° N, 71.8749° E	4.0	96
09	North	Kohat	33.5889° N, 71.4429° E	3.6	86.4
10	North	Nowshera	34.0105° N, 71.9876° E	3.6	86.4
11	North	Peshawar	34.0151° N, 71.5249° E	3.6	86.4
12	North	Abbottabad	34.1688° N, 73.2215° E	3.7	88.8
13	North	Gilgit	35.8819° N, 74.4643° E	2.3	55.2
14	North	Mingora	34.7717° N, 72.3602° E	3.4	81.6
15	North	Mirpur	33.1480° N, 73.7537° E	2.0	48
16	North	Muzaffarabad	34.3551° N, 73.4769° E	5.3	127.2
17	Central	Lahore	31.5204° N, 74.3587° E	5.8	139.2
18	Central	Sheikhupura	31.7167° N, 73.9850° E	4.1	98.4
19	Central	Bhakkar	31.6082° N, 71.0854° E	3.3	79.2
20	Central	Khushab	32.2955° N, 72.3489° E	2.0	48
21	Central	Mianwali	32.5839° N, 71.5370° E	2.8	67.2
22	Central	DG Khan	30.0489° N, 70.6455° E	2.0	48
23	Central	Layyah	30.9693° N, 70.9428° E	2.0	48
24	South	Karachi-I	24.9148° N, 66.8888° E	8.6	206.4
25	South	Karachi-II	24.8458° N, 66.9794° E	3.7	88.8
26	South	Badin	24.6459° N, 68.8467° E	2.6	62.4
27	South	Hyderabad	17.3850° N, 78.4867° E	3.4	81.6
28	South	Mirpur Khas	25.5065° N, 69.0136° E	2.0	48
29	South	Ghotki	28.0271° N, 69.3235° E	3.4	81.6
30	South	Rajan Pur	29.1044° N, 70.3301° E	8.6	206.4
31	South	Rahim Yar Khan	28.4212° N, 70.2989° E	2.9	69.6
32	South	Sukkur	27.7244° N, 68.8228° E	2.0	48
33	South	Gawadar	25.1313° N, 62.3250° E	2.2	52.8
34	South	Quetta	30.1798° N, 66.9750° E	3.1	74.4

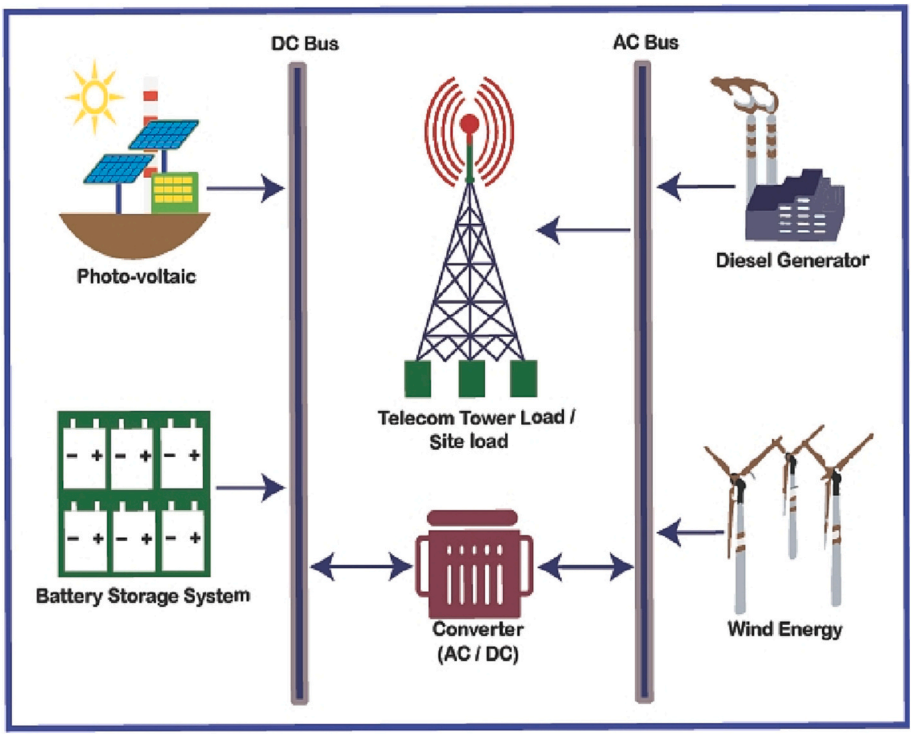


Fig. 4. Proposed Base Transceiver Stations (BTS).

$$P_{BP} = C_{in} + \sum_{t=1}^{T} \frac{1}{(1+r)^t} (C_{an}(t)^* A^* B_{an}) \times T \tag{8}$$

$$IRR(\%) = \frac{1}{(1+r)^t} \left(cashflow - \left(I_{inv} \times (1+r)^t \right) \right) \times 100 \tag{9}$$

$$ROI(\%) = \frac{1}{R} \left(\frac{\sum_{i=0}^{R} C_{i,ref}}{C_{cap} - C_{cap,ref}} - \frac{C_i}{C_{cap} - C_{cap,ref}} \right) \tag{10}$$

In financial analysis, the internal rate of return (IRR) is a statistic used to calculate the profitability of possible investments. In a discounted cash flow analysis, IRR is a discount rate that reduces all cash flows' net present values to zero, mathematical formula is shown in Eq. (9). Return on Investment (ROI) is the proportion of the average yearly change in nominal cash flows to the change in capital costs is shown in Eq. (10). It represents a cost saving over the long term compared to the initial investment.

The CF parameter shown in (11), the fraction of the standard control with the most remarkable strength for the theoretical yield, is another factor affecting performance evaluation. $E_{r,elect}$ and $E_{a,yr}$ are rated as the actual electrical energy of the system respectively. According to mathematics, CF is the system's rated capacity proportion to the average electrical power produced, as shown in Eq. (12).

$$CF_A(\%) = \left(\frac{E_{a,yr}}{E_{r,elec} \times 365 \times 24} \right) \times 100 \tag{11}$$

$$\eta_{sys}(\%) = [(\eta_{PV} + \eta_{wt} + \eta_B) \times \eta_{conv}] \times 100 \tag{12}$$

The mathematical formula of system efficiency is expressed in Eq. (12), which is the sum of all RESs efficiencies that are included in the system.

3.2. Site selection and load assessment

One of the most expensive expenses for telecommunications companies is energy consumption. The base transceiver station is one of the main components of cell sites that consume energy. Diesel fuel purchases for generators, which make up over 80 % of plant-level energy expenditures at off-grid and off-grid tower sites, are the primary source of these costs. There are thousands of BTS sites in Pakistan expanding in all three regions (North, South, and Central). In this study, there are 35 BTS sites are taken for the implementation of a hybrid renewable energy system shown in Fig. 1. The load assessment for all the BTS selected sites is taken from Pakistan Telecommunication Company (PLC). The detailed load assessment of all BTS sites is shown in Fig. 3 and Table 3.

3.3. Existing and proposed Base Transceiver Stations (BTS) design framework

The only electrical source currently in service in the Base Transceiver Stations (BTS) is a diesel generator. As a result, diesel generators are not economical and are not environmentally friendly. Therefore, these sites must integrate sustainable energy sources like wind and solar [4]. The finding suggests that various readily accessible renewable energy sources, including wind, tidal, ocean, and solar energy, are commonly utilized. As a result, because they are clean, green, and ecologically beneficial, renewable energy sources have replaced conventional ones (such as diesel, etc.) [56,57]. The installation has doubled due to the progressive development in hybrid renewable energy integration over the previous five years. The technology of the modern world favors distributed energy production. The proposed BTS model is shown in Fig. 4.

4. System modeling

According to IEEE, a microgrid is described as [58]; MG is a collection of interconnected loads, distributed energy sources, such as solar, wind, biomass, tidal, and geothermal, and energy storage technologies, such as batteries and hydrogen storage that function as a single as well as controllable unit with the grid.

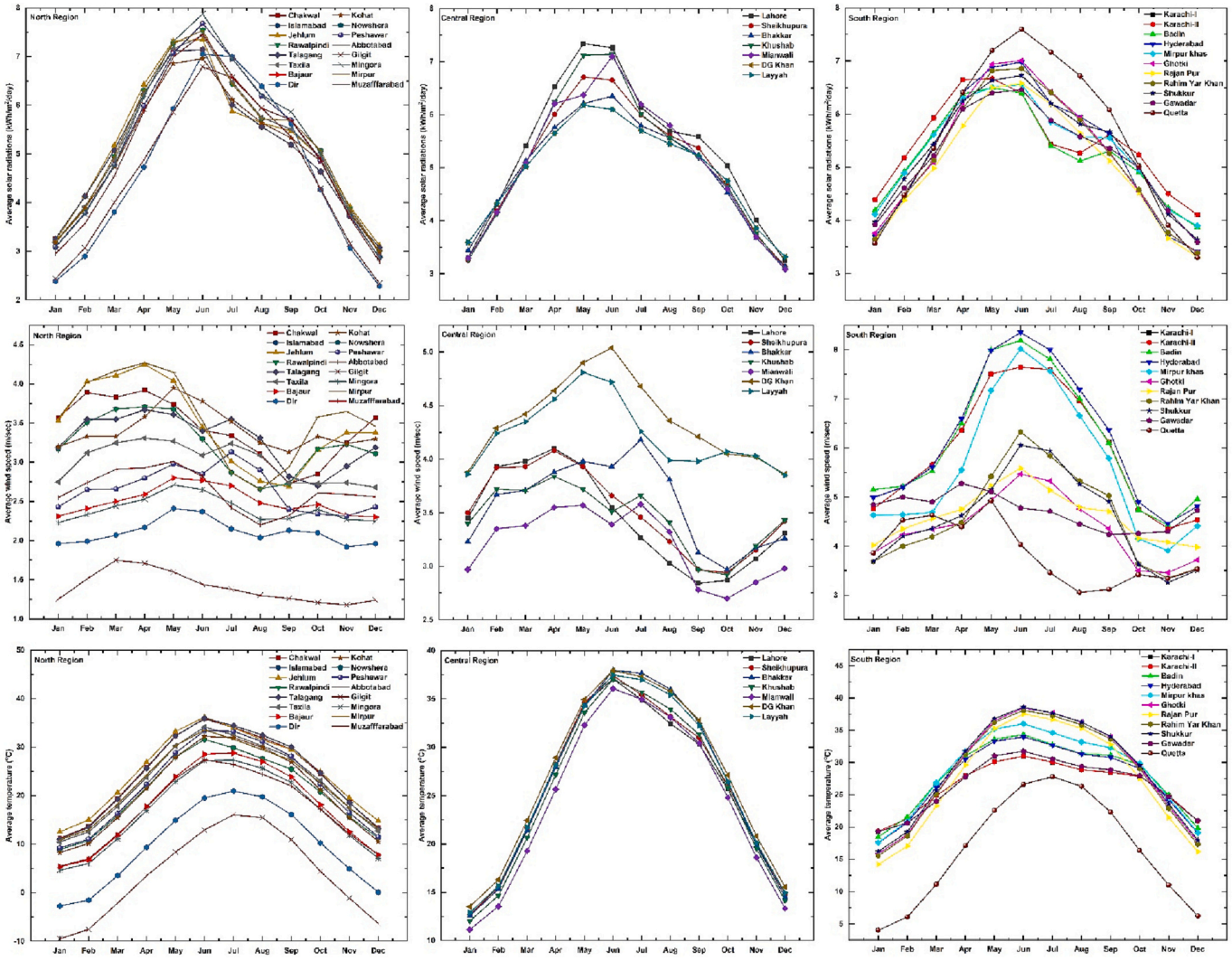


Fig. 5. Monthly average understudy BTS site date; solar irradiance (top), wind speed (middle), and temperature (bottom).

4.1. System components and resource assessment

In this proposed research, the modified BTS sites contain hybrid renewable energy, including solar, wind, and diesel generator, for backup to get sustainability with the battery storage system.

Therefore, the detailed modeling of each component included in this model and its corresponding resource assessment used in the BTS model are summarized below.

4.1.1. Photovoltaic system

One of the critical criteria for choosing a location for installing PV systems is GHI. GHI combines atmospheric diffuse horizontal and direct normal irradiance (DNI). The hourly profile depending on a NASA built-in database is utilized in the proposed study because on-ground data is not available for each of the chosen sites.

Eq. (13) calculated the clearance index and synthesized the results to hourly data based on the GHI profile using a built-in HOMER Pro [12]. In Eq. (13), the $H_{y,avg}$ is solar radiation (kWh/m^2/day) at top of the earth's atmosphere with $H_{m,avg}$ is the monthly average solar radiation at the top of the horizontal surface of the earth's atmosphere. The monthly average solar radiation profile of the nominated BTS sites is presented in Fig. 5 (top). It can be seen from Fig. 5 (top), that in the central region of Pakistan, Lahore had maximum average solar irradiance in June that is 7.26 (kWh/m^2/day), and Layyah had minimum average solar irradiance in June that is 6.18 (kWh/m^2/day).

In the northern region of Pakistan, Mingora had maximum solar irradiance in June, which is 7.88 (kWh/m^2/day). On the other hand, Gilgit has minimum solar irradiance, even in June, which is 6.79 (kWh/m^2/day). At last, the southern region of Pakistan has slightly lesser average solar irradiance than the north and central. In the southern area, in June, Quetta had maximum average solar irradiance of 7.6 (kWh/m^2/day), while Karachi had minimum solar irradiance in June, which was 6.4 (kWh/m^2/day). A flat-plate LONGi Solar LR6-72PE solar module is used in this study. It utilizes a 370Wp rated capacity with a fixed optimum slope and azimuth. This PV module was selected because of its superior efficiency, easy accessibility to the local market, cost-effectiveness, and suitability for sites with limited space. The cost and respective technical parameters of the selected PV segment (LONGi Solar LR6-72PE) are enlisted in Table 4 [12,54].

The PV module's output electrical power purely depends upon solar radiation. Therefore, the PV module generates the output power by using Eq. (14). In addition, in this study, to model the derating factor,

Table 4
Cost ($) with technical parameters of selected solar module (LONGi Solar LR6-72PE).

Capital cost	O&M cost	Replacement cost	Efficiency	Operating temperature	Rated capacity	Temp. coefficient	Lifetime	Derating factor	Ground reflectance
($)	($/year)	($)	(%)	(°C)	kWp	(%/°C)	(year)	(%)	(%)
259	3.70	185	19.1	47	0.370	−0.38	25.00	80.00	20.00

$$K_T = \frac{H_{m,avg}}{H_{y,avg}} \& H_{y,avg} = \frac{\sum_{n=1}^{365} G_o}{365} \tag{13}$$

the temperature effect on the PV module is also added.

$$P_{PV}(t) = \left[1 + \tau_{PV}\left(T_c - T_{c,STC}\right)\right] \times P_r D_{PV} \times \frac{H_T}{H_{T,STC}} \tag{14}$$

$$\tau_{PV} = \frac{V_{OC}}{V_{np}} \tag{15}$$

$$v\theta H_T = H_T \frac{h_t(T_c - T_a)}{H_T} + \eta_c \tag{16}$$

In Eq. (14), PV module rated power that is given by the manufacture is denoted by P_r, the derating factor is denoted by the D_{PV}, total solar radiations at standard temperature and pressure is denoted *by* $H_{T,\,STC}$, H_T is known for total solar irradiance, t_{pv} is known as the temperature coefficient of the PV panel with value $(-3.7{*}10^{-3})/^{o}$C, $T_{c,STC,}$ and T_c is called temperature (°C) of the cell at STC.

$$T_c = T_z + \frac{H_T(k\theta)}{h_t}\left(1 - \frac{\eta_c}{v\theta}\right) \tag{17}$$

$$\frac{k\theta}{h_t} = \frac{T_{C,NOCT} - T_{z,NOCT}}{H_{T,NOCT}} \tag{18}$$

$$T_C = H_T\left(1 - \frac{\eta_c}{k\theta}\right)\left(\frac{T_{C,NOCT} - T_{z,NOCT}}{H_{T,NOCT}}\right) + T_a \tag{19}$$

$$\eta_{mppt} = \left[1 + \tau_{PV}\left(T_C - T_{C,STC}\right)\right] \times \eta_{mppt,STC} \tag{20}$$

By utilizing solar transmittance (k) and absorptance (θ) measurements, HOMER Pro adopts the Duffie and Beckman technique [59] to determine the temperature of the cell (T_c), following correlations Eq. (16) and Eq. (20). Where T_z and $T_{z,\,NOCT}$ denotes the ambient temperature (°C) and STC conditions respectively. Therefore, the incident temperature at NOCT is 0.8 kW/m^2 is denoted by $H_{T,\,NOCT}$ and PV array electrical conversion efficiency and MPPT is denoted by η_c with η_{mppt} as the efficiency of maximum power point tracking.

4.1.2. Wind turbine

Wind speed is supposed to influence the choice and placement of wind turbines. Wind turbines convert mechanical energy from wind speed into mechanical energy, which is converted into electrical energy. HOMER Pro uses the built-in profile provided by NASA to conduct the suggested investigation because there is a paucity of on-ground data. Based on the Power Law as Eq. (21) [60], HOMER Pro determines the scaled wind speed data from the baseline data following the various hub heights of the chosen wind turbines.

$$\mu_{hh} = \mu_{ane}\left(\frac{H}{H_o}\right)^R \tag{21}$$

In Eq. (21), μ_{ane} and μ_{hh} are the speed of wind at anemometer and hug height respectively. While H_o and H are the heights of the anemometer and hub respectively. In addition, R is the roughness of the surface having a value of 0.10. To model the wind system in HOMER Pro, there are necessary parameters that are being called the Weibull distribution function considered as shown in Eq. (22) and (23).

$$f(v) = exp\left(-\frac{v}{c}\right)^k * \frac{k}{c}\left(\frac{v}{c}\right)^{k-1} \tag{22}$$

$$v = c\varphi\left(1 + \frac{1}{k}\right) \tag{23}$$

where, the parameters v, k, and c are wind speed (m/s), Weibull shape, and Weibull scale respectively. However, Φ represents the gamma function. So, the average monthly wind speed of the nominated BTS sites is displayed in Fig. 5 (middle). The southern region of Pakistan has a high average wind speed of 8.0 m/s rather to the north and central regions of Pakistan.

In this study, there are four wind turbines model, including Bergey Excel 6-R (6 kW), Bergey Excel 10 (10 kW), Ecocycle E010 (10 kW), and Xzeres 7.2 (10 kW). Their power curves are shown respectively in Fig. 6. Finally, Bergey Excel 10 (10 kW) was considered the optimal wind turbine based on parameters including high cutoff speed, i.e., 20 m/s, low cut-in wind speed, i.e., 3 m/s and also additional benefits of easy transportation and installation. The cost with technical parameters and losses associated with a wind turbine is shown in Table 5 [12,54]. The mathematical expression for power density (E) is described in Eq. (24) while, to get the wind output power, the formula used is shown in Eq. (25).

It can be observed that output power purely depends upon the air density (ρ), speed of wind (v), and cross-section area (A) at STP.

$$E\left(\frac{w}{m^2}\right) = \vartheta\left(1 + \frac{3}{k}\right) \times c^3 \times f(v) \tag{24}$$

$$P_{wind} = \frac{1}{2}\rho A V^3 \tag{25}$$

$$P_{WT} = P_{WT,STP} \times \left(\frac{\rho}{\rho_d}\right) \tag{26}$$

In Eq. (26), ρ_d is air density having a value of 1.225 kg/m3, and ρ is actual air density. The air density and wind turbine output power being affected by the altitude. Therefore, air density and its ratio are calculated by Eqs. (27) and (28) [59].

$$\rho = \frac{P}{RT} \tag{27}$$

$$\frac{\rho}{\rho_O} = \left(\frac{T_O}{T_O - Bz}\right) \times \left(\frac{T_O - Bz}{T_O}\right)^{\left(\frac{g}{RB}\right)} \tag{28}$$

In Eqs. (27), pressure is denoted by P (Pa), and the temperature in kelvin is denoted by T, gas constant (R) having a value of 287 J/kgK. While determining the wind turbine output power at a particular altitude, HOMER uses the ratio of air density, which is defined as the ratio between the air density under actual and ideal situations.

Hence, B is called the lapse rate (0.00650 K/m), T_o is the ambient temperature (288.16 K), Z is called the altitude (m), and g is the gravitational acceleration (9.81 m/s2). Wind turbine power chrematistics is expressed in Eq. (30).

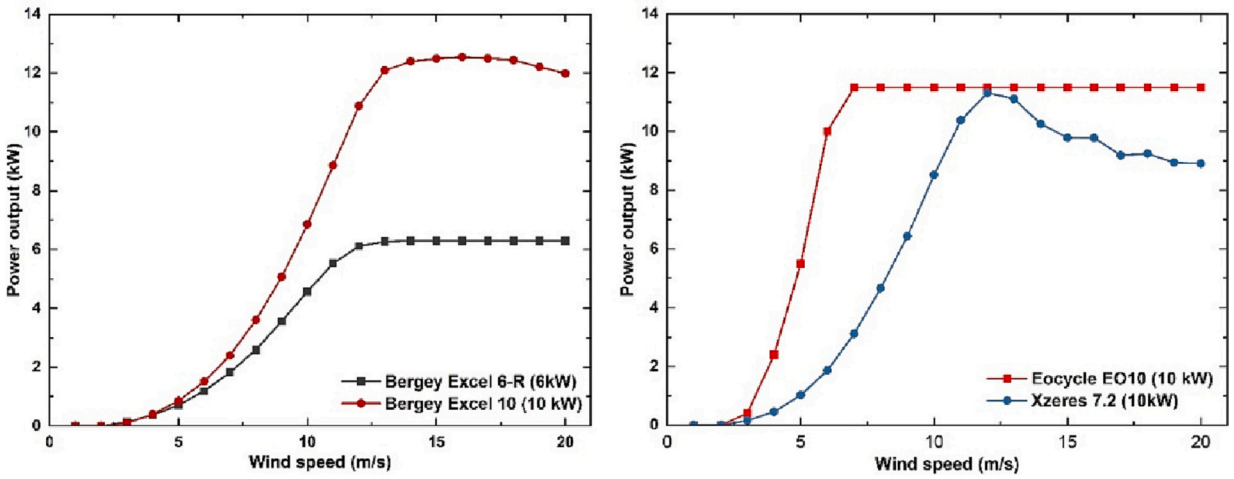


Fig. 6. Power curves for four evaluated wind turbines.

Table 5
Cost with technical parameters and losses of selected wind turbine (Bergey Excel 10 (10 kW)).

Capital cost	O&M cost	Replacement cost	Rated capacity	(T_a) effect	Rated capacity	Temp. coefficient	Lifetime
($/kW)	($/year)	($/kW/year)	(kW)		kW	(%/°C)	(year)
1500	1250	100	10	Yes	0.370	−0.38	20.00

Loss type	Availability loss	Wake effect loss	Electrical loss	Balance of plant loss	Miscellaneous loss	Overall loss factor
	(%)	(%)	(%)	(%)	(%)	(%)
Value	5.18	5.0	10	1.7	10.54	21.02702

$$P_{WT,STP} = C_P \times \frac{1}{2}\rho A v^3 \tag{29}$$

$$\mathrm{P_{rated}} = \mathrm{P_{WT}} \text{ if } \mathrm{v_{cut-out}} \geq \mathrm{v} \geq \mathrm{v_{rated}}$$

$$\mathrm{P_{WT}} = \mathrm{P_{rated}}\left(\frac{\mathrm{v}^3}{\mathrm{v^3_{rated}} - \mathrm{v^3_{cut-in}}}\right) - \mathrm{v^3_{cut-in}}\left(\frac{\mathrm{P_{rated}}}{\mathrm{v^3_{rated}} - \mathrm{v^3_{cut-in}}}\right) \text{ if } \mathrm{v_{rated}} \geq \mathrm{v} \geq \mathrm{v_{cut-in}}$$

$$P_{WT} = 0 \; if \; v_{cut-in} \geq v \geq v_{cut-out} \tag{30}$$

$$L_{overall} = \sum_{i=1}^{n} 1 + (L_i/100) \tag{31}$$

It is important to note that the numerous losses connected with wind turbines are also considered in this analysis to reflect realistic setups. These losses are gathered through multiple licenses given by NEPRA to public and private businesses to develop wind farms with various capacities, described in Table 5 [54]. Environmental (temperature) losses, turbine performance losses, and other erroneous losses are examples. In Eq. (31), *Li* denotes the specific losses connected to each type, while $L_{overall}$ is the overall loss factor.

4.1.3. Battery storage system

This study uses the generic lithium-ion (G-LI-ASM) progressive battery storage model to provide the best performance given the climatic (temperature) characteristics of the under-studied areas. In generic lithium-ion, the temperature effects are taken to simulate the deprivation losses. Since a PV module with a nominal rated voltage of 40 V is connected to a DC bus as an energy source [54], to retain 40 V at the DC bus, the string size for batteries is fixed at 11. Table 6 [54,61] illustrates both cost and other technical specifications.

The storage capacity is expressed by a mathematical formula in Eq. (32), where total demand of energy is denoted by E_{TD}, *DDB* is denoted for depth discharge of the battery, B_B is known for battery autonomy that is expressed in Eq.(33), η_{bat} and η_{conv} are converters and battery efficiency respectively, E_{nom} is called battery nominal capacity and measured in Ah, while battery minimum state of charge is denoted by SOC_{min} (%) and $P_{avg,prime,load}$ is called average primary load (kWh/day).

$$E_{cap}(t) = \eta_{bat} \times (E_{TD} \times B_B)\eta_{conv} \times DDB \tag{32}$$

$$B_B = \frac{\frac{24h}{d} \times N_B \times \left(1 - \frac{SOC_{min}}{100}\right) \times V_{nom} \times E_{nom}}{P_{avg,prim,load}\left(\frac{1000Wh}{kWh}\right)} \tag{33}$$

Therefore, the charging and discharging of the battery are formulated in Eqs. (34) and (35), where $P^{+(t)}$ and $P^{-(t)}$ demonstrate charging and discharging of the battery at a specific period "*t*", load demand is denoted by $P_{load}(t)$. In contrast, $E_{min}(t)$ and $E_{max}(t)$ represent electrical energy storage capacity at a specific period "*t*" respectively.

Table 6
Cost with technical parameters of selected battery storage model (Generic 1kWh Li-ion [ASM]).

Capital cost	O&M cost	Replacement cost	Nominal capacity	Minimum state of charge	Initial state of charge	Degradation limit	Trip losses	Ground reflectance
($)	($)	($)	(kWh)	(%)	(%)	(%)	(%)	(%)
200	10.00	200	1.02	20.00	100	30.00	8.00	20.00

$$P^{+(t)} if\ P_{load(t)} < P_{gen(t)} \& E_{max(t)} \geq E_{(t)} \tag{34}$$

$$P^{-(t)} if\ P_{load(t)} > P_{gen(t)} \& E_{max(t)} \geq E_{(t)} \geq E_{min(t)} \tag{35}$$

The Eq. (34) and Eq. (35) ensure that both charging and discharging are not simultaneously occurring at the same time. Once the generated power by renewable energy sources $P_{gen}(t)$ is greater than the load demand $P_{load}(t)$ of the system, the battery charging cycle activate to charge the battery and to absorb the excess generated power. But at the same time, these constraints also ensure the battery upper and lower SOC limits. Charging (Eq. (34)) will not be processed if the SOC of battery reaches upper limits. Similarly, at the same time discharging (Eq. (35)) cannot be processed if the SOC of battery is below the lower limit (20 %). This can prevent battery degradation and save the charging/discharging cycle of the battery.

The amount of energy $E_{(t)}$ currently available in the battery bank at time slot "t" constantly equals the total amount of previously available energy from period $E(t\text{-}1)$ and recent time slot t. The given Eq. (36) can be used to express the power available in the battery bank at the specified time "t". Where the terms η_{ch} and $\eta_{b,ch}$ stands for the respective charge controller and efficiencies of battery charging respectively. The formula used for the amount of state of charge of the battery (SOC) and depth of discharge (DOD) shown in Eq. (37) to Eq. (39).

$$E_{b(t)} = E_{(t-1)} + E_{(t)} \times \eta_{b,ch} \times \eta_{ch} \tag{36}$$

$$SOC_{max(t)} \geq SOC_{(t)} \geq SOC_{min(t)} \tag{37}$$

$$SOC_{min(t)} = 1 - DDB_{(t)} \tag{38}$$

$$DDB_{max(t)} \geq DDB_{(t)} \geq DDB_{min(t)} \tag{39}$$

Eqs. (40) and (41) [59] illustrate the quantity of charged and discharged power through a battery for a precise time. Where $E(t)$ is the bound energy at the start of the time step, maximum storage capacity is denoted by $Emax$, c is the ratio of storage capacity, and k is the constant associated with conductance.

$$P^{+}{}_{max} = \frac{k\left[\left(E_{(t)} \times c(1 - e^{-k\Delta t})\right) + (E(t-1)e^{-k\Delta t}) + \left(-c \times E_{max(t)}\right)\right]}{1 + c(k\Delta t + e^{-k\Delta t} - 1) - e^{-k\Delta t}} \tag{40}$$

$$P^{-}{}_{max} = \frac{(kEte^{-k\Delta t} - kEe^{-k\Delta t}) + \left(E_{(t)}kc\right) - \left(E_{(t)}kc^{*}e^{-k\Delta t}\right)}{1 + c - e^{-k\Delta t} + ck\Delta t + ce^{-k\Delta t}} \tag{41}$$

As shown in Eqs. (42) to (45), HOMER Pro determines the battery bank's maximum power by imposing three different storage constraints [11], where the kinetic storage model is the source of the first constraint. The second deals with a storage system's maximum charge rate, while the third deals with a storage system's maximum charge current. Where α_c is called the maximum storage charge rate (A/Ah), the maximum storage charge current is denoted by I_{max}, the total number of batteries present in the storage bank is denoted by Nb, and the nominal voltage level is denoted by V_{nom}.

$$P^{-}{}_{max,aa} = \frac{(kE(t-1)e^{-k\Delta t}) + \left(-kc \times E_{max(t)} \times E_{(t)}\right)}{1 + c - e^{-k\Delta t} + ck\Delta t + ce^{-k\Delta t}} \tag{42}$$

$$P^{-}{}_{max,bb} = \frac{\left(E_{max(t)} \times E_{(t)}\right) \times (1 - e^{-\propto_c \Delta t})}{\Delta t} \tag{43}$$

$$P^{-}{}_{max,cc} = \frac{N_b\, I_{max} V_{nom}}{1000} \tag{44}$$

$$P^{-}{}_{max} = \frac{min\left(P^{-}{}_{max,aa}, P^{-}{}_{max,bb}, P^{-}{}_{max,cc}\right)}{\eta_{b,ch}} \tag{45}$$

$$P_{out} = I(V_{nom} - IR_s) \tag{46}$$

$$I = \frac{V_{nom}}{2R_s} \tag{47}$$

The effects of temperature on battery deterioration are considered. As a result, the thermal model will be used to simulate how a specific temperature affects battery performance. Thus, a revised kinetic model of a Li-ion battery is used in this study's advanced model, and the losses are approximated using a series resistance (Rs). Eqs. (46) and (47) calculate the output power (Pout) for a given applied current (I).

Where output power is denoted by P_{out} with nominal voltage as V_{nom} that is converted to heat as shown in Eqs. (48) and (49). Where thermal conductance is denoted by the "h", specific heat (J/kgk) is denoted by the "c", mass of battery is denoted by "m", "T_b" is called Kelvin, internal and ambient temperature is denoted by "T" and "T_a". Therefore, the storage bank temperature in the next period time slot is denoted by "T_{i+1}" which is shown in Eq. (50).

$$Q_h = I^2 R_s \tag{48}$$

$$m_c T_b = Q_h - \frac{1}{h}[T - T_a] \tag{49}$$

$$T_{i+1} = \frac{Q_h}{h} + T_a\left(1 - e^{\left(-\frac{h}{mc}\right)dt}\right) + \left(T_i - \frac{Q_h}{h}\right) \times e^{\left(-\frac{h}{mc}\right)dt} \tag{50}$$

$$kt = C \times e^{\left(-\frac{d}{T}\right)} \tag{51}$$

According to Eq. (51), this variable's growth rate solely depends on temperature. Where time and temperature degradation variable's rate of increase is denoted by "kt". "b" and "d" are variables.

4.1.4. Power converter

An electrical or electromechanical device used to convert electrical energy is called a power converter. Altering current (AC) into direct current (DC), changing the voltage or frequency of the current, or any combination of these are all possible using a power converter. A bi-directional power converter is connected chiefly between the AC and DC buses and works in tandem with the AC generator for two-way power conversions. Therefore, the converter rating is usually chosen after the power rating of the energy source linked to the appropriate bus, with an expected converter efficiency of 85 %.

$$C_{p.conv}(kW) = P_{array} \times \frac{100}{85} \tag{52}$$

$$\eta_{p.conv}(\%) = \frac{Output_{conv}}{Input_{conv}} \times 100 \tag{53}$$

Since solar PV is linked in this instance as a source of energy at the DC bus. Hence, converter efficiency and capacity are chosen by Eq. (52) and (53). Therefore, technical parameters with the cost of the designated converter are shown in Table 7 [54].

4.2. System design constraints

In this study, the fundamental model constraint is the energy balance, which states that the total power produced by all RESs must meet the base load demand at generated energy for each time period. Eq. (54) represents the immediate restriction on energy balance mathematically.

Table 7
Cost with technical parameters of the selected converter.

Capital cost	Replacement cost	Lifetime	Efficiency	Relative capacity
($/kW)	($/kWh)	(years)	(%)	(%)
150	150	15	95	100

$$\sum_{t=1}^{T}\left(P_{gen,PV}+P_{gen,WT}\right)=\sum_{t=1}^{T}\left(P_{load}+P_{loss}+P_{BES}+P_{excess}\right) \quad (54)$$

$$P_{gen,max(t)} \geq P_{gen(t)}+P_{gen,min(t)} \quad (55)$$

$$\sum_{t}^{T}\left(P_{gen,(t)}\right)=\left(P_{load}+P_{loss}+P_{BES}+P_{excess}\right) \quad (56)$$

The total generated power for a given period t must fall between generation limits, just like in Eq. (55). Similar to Eq. (56), the power generated ($P_{gen(t)}$) should equal the sum of the power used to serve the primary AC load (P_{load}), the total power lost (P_{loss}), battery energy storage (P_{BES}), and the extra power (P_{excess}) that is still available after the time period.

5. Results analysis and discussion

Four partitions are utilized to carry out the analysis and discussions of the findings. The first partition includes the conclusions and discussion of the techno-economic analysis based on objective functions for all designated BTS sites; the second partition compares the existing and proposed BTS sites. In the third fold, a GHG emissions-based environmental analysis is conducted. In the fourth partition, a sensitivity analysis based on numerous uncertainties is undertaken, including the discount rate, inflation rate, solar irradiation, wind speed, local load demand, etc.

5.1. Techno-economic optimization

Techno-economic optimization aims to size system components as efficiently as possible without sacrificing the system's affordability or dependability. HOMER Pro simulates 34 different BTS sites while considering site-specific factors. Depending on NPC and LCOE, the suggested optimal BTS sites are tailored to each unique area. Table 8 lists all BTS sites with associated optimal system formations and decision parameters in this initial analysis. At the chosen site, the possibility for diverse energy resource availability and regional primary electricity demand are both crucial factors in determining the best system layout, component sizing, and cost parameters. To quantify the analysis, an ideal BTS site configuration category is chosen for each understudied location. Sections 5.1.1 and 5.2.2 examine the techno-economic optimization for each BTS site configuration with specific component sizing and other characteristics. Therefore, Fig. 7, and Fig. 8, illustrate a comparative analysis based on NPC, LCOE, PBP, excess electricity, IRR, and ROI for hybrid setups of more than two sources using a radar chart.

Table 8
Optimal configuration for understudy BTS sites with final decision indicators.

BTS site no.	Project area	Optimal BTS configuration	Dispatch strategy	Cost and objective parameters			Financial parameters			Capacity shortage	Excess energy	Unmeet load
				LCOE	OPC	NPC	IRR	ROI	PBP			
				($/kWh)	($/yr.)	($M)	(%)	(%)	(yr.)	(%)	(%)	(%)
Northern zone												
BTS-01	Chakwal	PV-B-DG	LF	0.13	1683	0.06892	24	20	4	0	31.8	0
BTS-02	Islamabad	PV-B-DG	LF	0.1268	4316	0.17289	26	21	3.8	0	31.4	0
BTS-03	Jehlum	PV-B-DG	LF	0.1296	1872	0.07609	25	20	3.9	0	29.2	0
BTS-04	Rawalpindi	PV-B-DG	LF	0.127	3685	0.14912	26	21	3.8	0	31.4	0
BTS-05	Talagang	PV-B-DG	LF	0.1306	1385	0.05687	24	20	4	0	32.8	0
BTS-06	Taxila	PV-B-DG	LF	0.1278	3573	0.14518	25	21	3.9	0	30.8	0
BTS-07	Bajaur	PV-B-DG	LF	0.1267	5104	0.20624	26	21	3.8	0	31.8	0
BTS-08	Dir	PV-B-DG	LF	0.1815	4226	0.13747	20	21.6	4.8	0	40.1	0
BTS-09	Kohat	PV-B-DG	LF	0.1322	2232	0.09013	24	20	4	0	32.9	0
BTS-10	Nowshera	PV-B-DG	LF	0.1296	2145	0.08837	24	20	4	0	32.8	0
BTS-11	Peshawar	PV-B-DG	LF	0.1297	2161	0.88415	25	20	4	0	32.8	0
BTS-12	Abbottabad	PV-B-DG	LF	0.1328	2321	0.09303	24	20	4	0	33.3	0
BTS-13	Gilgit	PV-B-DG	LF	0.2325	3511	0.10124	17	13	5.7	0	44.6	0
BTS-14	Mingora	PV-B-DG	LF	0.1258	2043	0.08098	26.6	21.9	3.72	0	31.0	0
BTS-15	Mirpur	PV-B-DG	LF	0.129	1233	0.04893	25.6	20.9	3.84	0	32.7	0
BTS-16	Muzaffarabad	PV-B-DG	LF	0.1318	3374	0.13229	25.2	20.6	3.9	0	32.3	0
Central zone												
BTS-17	Lahore	PV-B-DG	LF	0.1272	3466	0.13964	26	21	3.8	0	31.2	0
BTS-18	Sheikhupura	PV-B-DG	LF	0.1343	2616	0.10423	24	20	4	0	30.0	0
BTS-19	Bhakkar	PV-B-DG	LF	0.1342	2120	0.08386	24	20	4	0	29.7	0
BTS-20	Khushab	PV-B-DG	LF	0.1327	1279	0.05026	25	20	3.9	0	32.4	0
BTS-21	Mianwali	PV-B-DG	LF	0.131	1711	0.06945	24	20	4	0	31.9	0
BTS-22	DG Khan	PV-B-DG	LF	0.1363	1313	0.05161	24	19	4.1	0	32.9	0
BTS-23	Layyah	PV-B-DG	LF	0.1348	1290	0.05103	24	20	4	0	33.1	0
Southern zone												
BTS-24	Karachi-I	PV-W-B-DG	LF	0.1096	4098	0.17839	28.5	24.1	3.5	0	31.3	0
BTS-25	Karachi-II	PV-W-B-DG	LF	0.1111	1754	0.07785	27.6	23.1	3.6	0	38.5	0
BTS-26	Badin	PV-W-B-DG	LF	0.118	1264	0.05807	25.1	20.6	3.92	0	48.8	0
BTS-27	Hyderabad	PV-W-B-DG	LF	0.1171	1664	0.07541	25.6	21.1	3.95	0	43.2	0
BTS-28	Mirpur Khas	PV-B-DG	LF	0.1303	1240	0.04932	25.1	20.5	3.9	0	31.2	0
BTS-29	Ghotki	PV-B-DG	LF	0.1333	2169	0.08580	24.7	20.1	3.97	0	30.7	0
BTS-30	Rajan Pur	PV-W-B-DG	LF	0.1294	4979	0.21096	23.8	19.4	4.13	0	30.9	0
BTS-31	Rahim Yar Khan	PV-B-DG	LF	0.1333	1840	0.07318	24.4	19.8	4.01	0	31.1	0
BTS-32	Sukkur	PV-B-DG	LF	0.1282	1244	0.04856	25.7	21.1	3.81	0	29.6	0
BTS-33	Gawadar	PV-B-DG	LF	0.1355	1380	0.05643	23.4	19	4.18	0	31.6	0
BTS-34	Quetta	PV-B-DG	LF	0.1246	1795	0.07314	26.1	21.7	3.77	0	28.6	0

Note: PV = Photovoltaic, W=Wind, DG = Diesel Generator, B = Battery, LCOE = Levelized cost of energy, LF = load following, NPC=Net present cost, OPC = Operating cost, ICC = Initial capital cost, IRR = Internal rate of return, ROI = Return on investment, PBP = Payback period, $M = million dollars

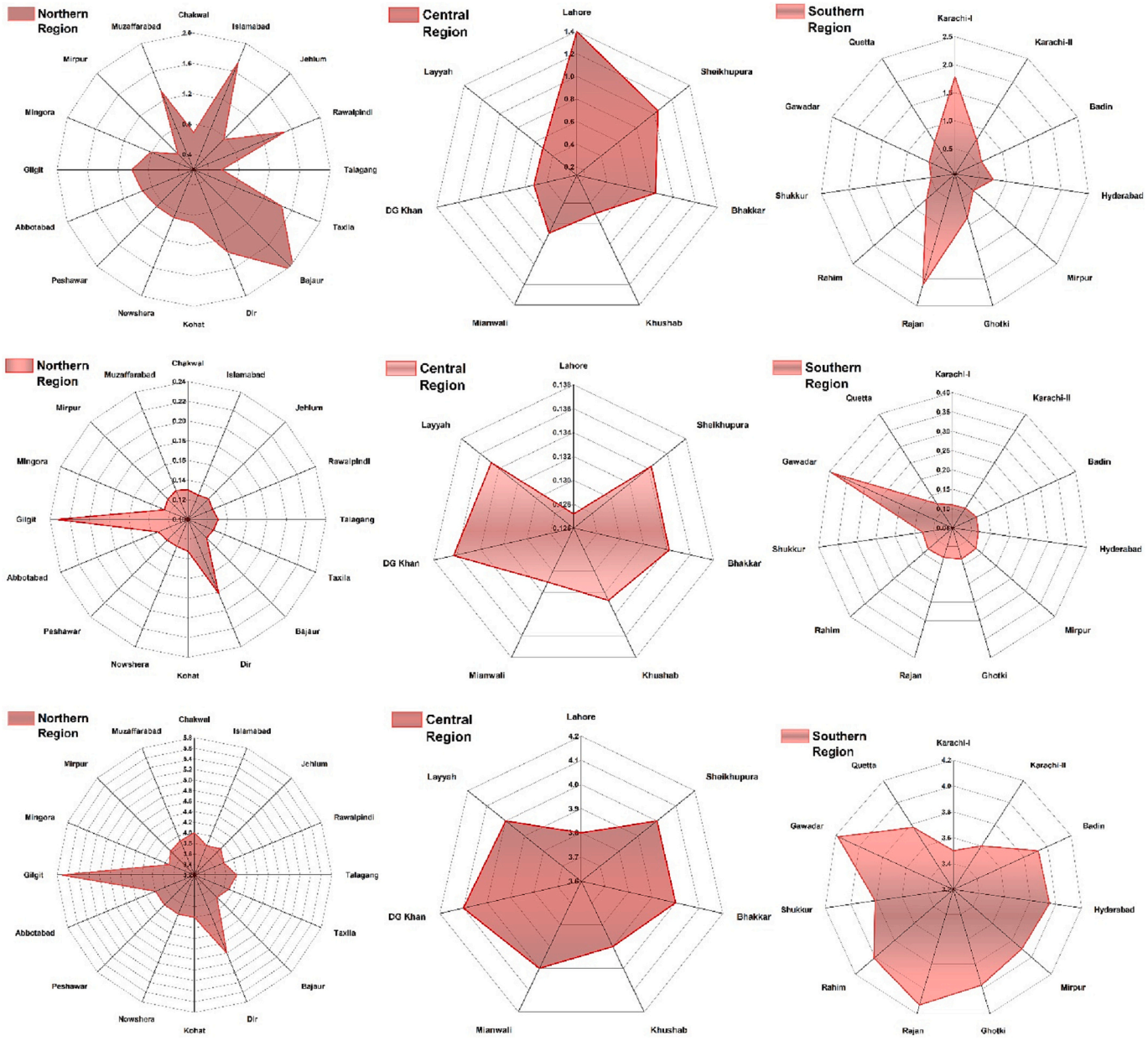


Fig. 7. Radar diagram for the Net present cost (lack dollars) (top), Levelized cost of energy ($/kWh) (middle), and Payback period (years) (bottom) of proposed optimal understudy BTS sites.

5.1.1. PV and DG with battery storage

In all optimization results, a total of 34 BTS sites were studied, in which 28 BTS sites reported PV-B-DG (PV-battery-diesel) based optimal configuration. It can be observed from Table 8, that all northern and central regions have PV-B-DG configurations because these regions have dominancy in solar energy with less wind energy. Therefore, the PV is the optimal solution for the battery storage system due to climate conditions in the north and central. Bajaur BTS-07 proved to be the best feasible site in the north region with the least LCOE of 0.1267 $/kWh and NPC of 0.206240$M. On the other side, Gilgit BTS-13 reported the highest LCOE in the north region of 0.2325 $/kWh and NPC of 0.10124 $M, due to the high electrical load and less solar radiation. The temperature difference is the second reason for the difference in LCOE in both comparative sites. The annual average temperature of Bajaur is 17.71 °C, while Gilgit's average temperature is 3.76 °C. Due to the low average temperature in Gilgit, especially in winter, solar irradiance is limited due to foggy days, which reduces PV efficiency and performance.

In contrast, in the central region, Lahore BTS-17 was ascertained to be the optimal site with the least 0.1272 $/kWh, with NPC 0.139648$M. At the same time, DG Khan BTS-22 reported with highest LCOE of 0.1363 $/kWh with NPC 0.05161$M. In the south region, the six BTS sites have PV-B-DG optimal configuration out of eleven sites. The Quetta BTS-34 with the least LCOE of 0.1246 $/kWh with NPC 0.07314$M due to higher solar radiation and temperature. While Gawadar BTS-33 with a maximum LCOE of 0.1355 $/kWh with NPC 0.05643$M. The comparative analysis of NPC, LCOE, PBP, Excess energy, IRR, and ROI of these under-discussed BTS sites are shown in Fig. 7 and Fig. 8.

5.1.2. PV-W and DG with battery storage

It is noted from Table 8, that only five sites that are in the southern region have the optimal configuration of PV-W-B-DG. Because it purely depends upon the wind speed. These five sites are Karachi-I BTS-24, Karachi-II BTS-25, Badin BTS-26, Hyderabad BTS-27, and Rajan Pur BTS-30. All these sites have maximum wind speed, as seen in Fig. 5

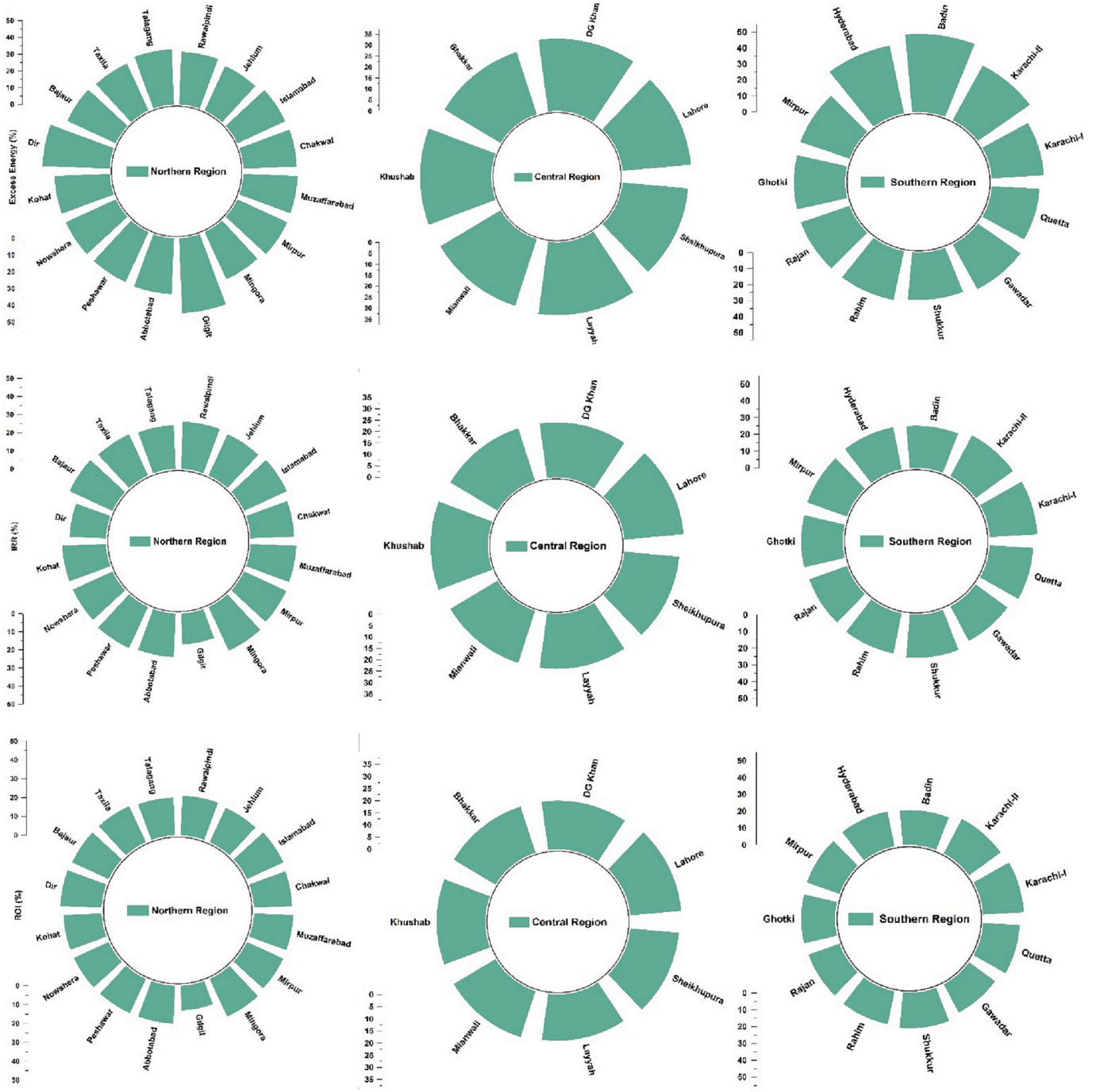


Fig. 8. Radar diagram for Excess energy (top), ROI (middle), and IRR (bottom) of proposed optimal understudy BTS sites.

(middle). In the southern region, Karachi-I BTS-24 has the least LCOE of 0.1096 $/kWh with NPC of 0.17839$M. Instead, Rajan Pur BTS-30 has a maximum LCOE of 0.1294 with an NPC of 0.210964$M. The difference in LCOE is due to the difference in wind speed of these two sites. The annual average wind speed of Karachi-I BTS-24 is 5.96 m/s, while Rajanpur BTS-30 has 4.61 m/s wind speed. The comparative analysis of NPC, LCOE, PBP, Excess energy, IRR, and ROI of these under-discussed BTS sites are shown in Fig. 7, and Fig. 8.

HOMER Pro uses two dispatch techniques. One is load-following dispatch, while the other is cycle charging (CC) dispatch. When the generator is called upon in CC dispatch, it runs at full capacity, and any additional energy is either logged as excess energy or used to charge the battery. When the penetration of renewable energy is low, this technique is likely to be the most effective. A generator is needed under a load-following dispatch method that only generates the required power. When several heterogeneous sources are introduced into the system, this

Table 9
Comparison between dispatch strategies following in this study.

Dispatch strategy	Levelized cost of energy (LCOE)	Excess electricity	Operating cost	Payback time
	($/kWh)	(%)	($)	(year)
Cycle Charging (CC)	0.3150	64.6	7282	8.94
Load Following (LF)	0.1246	28.6	1795	3.77

technique is shown to be the most effective when there is a high penetration of renewable energy [62]. Hence, the Quetta BTS-34 site is taken to compare the performance of both dispatch strategies illustrated in Table 9. In this scenario, the load following (LF) technique displayed a higher index than CC. These advantages of the load following (LF) method have been observed, and they are a result of its improved asset management standards and power regulation to satisfy load demand.

5.2. Comparative analysis of proposed and existing system

5.2.1. Levelized cost of energy

The results of this research study are compared based on the LCOE of the existing and proposed BTS site scenarios shown in Fig. 9. The existing infrastructure of BTS sites purely depends upon the diesel generator. Therefore, the existing BTS sites have higher LCOE due to diesel fuel prices. It can be observed that existing BTS sites have higher LCOE ranges from 0.377$/kWh to 0.3920 $/kWh. After integrating wind and PV sources with the BTS sites, all sites were noted as having lower LCOE, which is our primary objective in this study, having the average LCOE of all sites is 0.164$/kWh. In addition, in the proposed BTS sites, PV-B-DG configuration sites have slightly higher LCOE than PV-W-B-DG configuration sites. Because the efficiency of wind energy is higher than solar, those sites that have wind higher dependency can be more feasible. It can be highlighted from research that the proposed study showed excellent performance with the economic parameters as well rather than existing BTS sites.

5.2.2. Operating cost and Net present cost

The operating cost and net present cost of the existing and proposed BTS sites are shown in Fig. 10. It is noted that the operating cost and net present cost reduce due to the integration of renewable energy with the BTS site. The major reason for a reduction in operating and net present costs is less dependency on a diesel generator. Therefore, due to the reduction of DG hours and fuel consumption, renewable energy became dominant for meeting the electrical demand. Due to the reduction of fuel consumption dependency, the net present cost reduces from an average of 0.35$M in existing BTS to an average of 0.1$M in the proposed BTS. Economically and technically, solar is far better than a diesel generator because the cost of the region with renewable energy is less than the cost of the site with a conventional diesel generator. A detailed explanation of the PV (kW), number of wind turbines, DG generation (kW), DG hours, and fuel consumption with carbon emission is shown in Table A1.

5.3. Environmental analysis

Sustainability assessments include three main pillars: environmental, social, and economic sustainability. Considering the GHG emissions from every BTS site, an environmental assessment with a life cycle (25 years) is carried out in this research. Only CO2 emissions are mainly taken into account in this study due to their most prominent impact on the total GHG emission factor. However, emission factor related to system simulation and fuel usage is used to determine carbon emissions. The mathematical formulation for carbon emission estimation is expressed in Eq. (57). Where the emission factor is denoted by the E_f, the activity rate by A_r, and overall emission reduction efficiency (%) is denoted by η_{ER}.

$$E = A_r \times \frac{1}{100}(1 - \eta_{ER}) \times E_f \tag{57}$$

The complete comparison of carbon emission of all proposed and existing BTS sites is discussed, and a graphical presentation is shown in Fig. 11. In the existing BTS sites, carbon emission is much higher due to

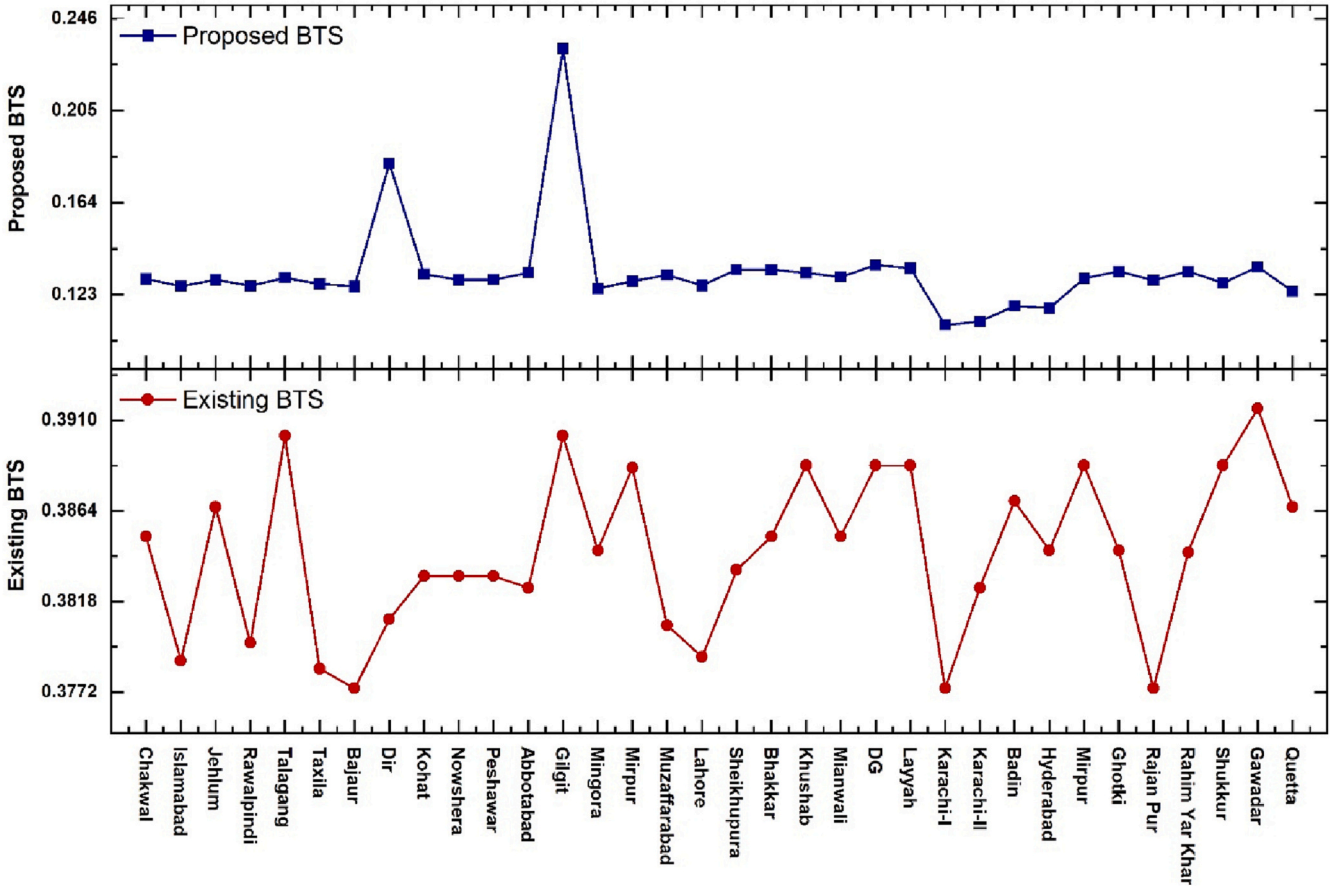


Fig. 9. Levelized cost of energy (LCOE) of existing and proposed BTS sites.

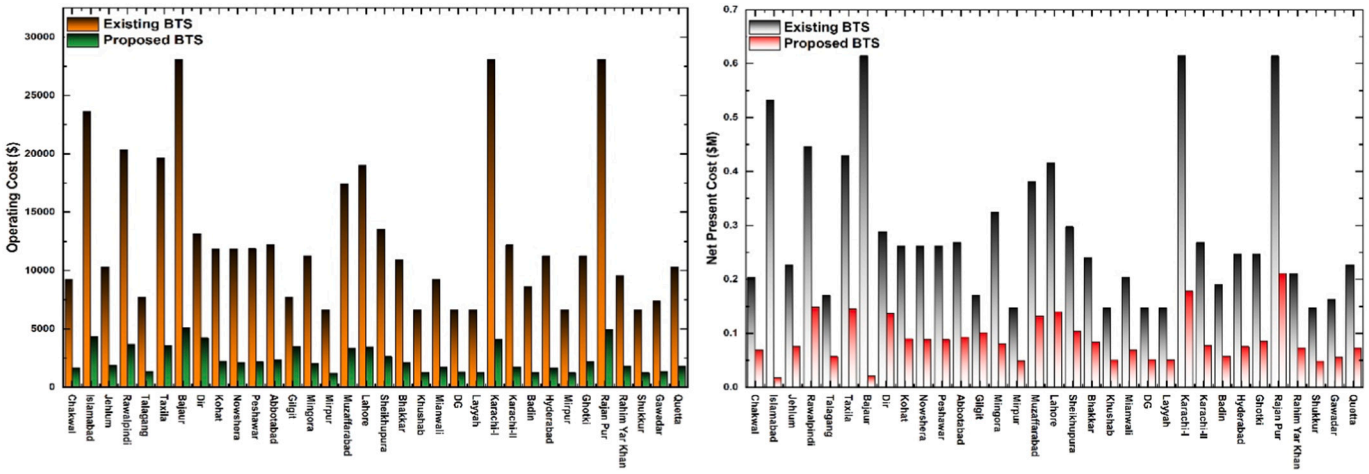


Fig. 10. Operating cost (OC) and Net present cost (NPC) of existing and proposed BTS sites.

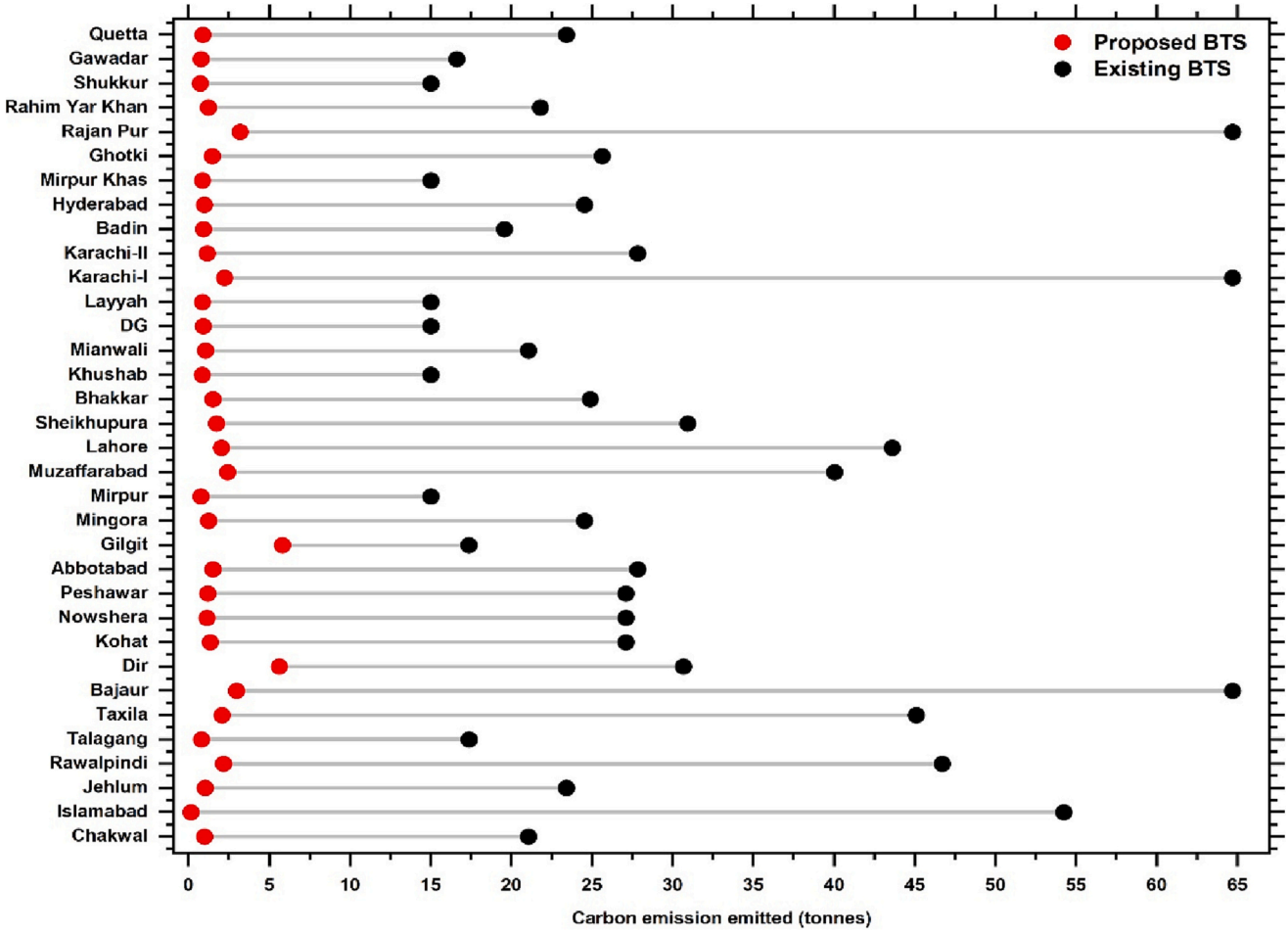


Fig. 11. Carbon emission (tones) analysis for all existing and proposed BTS sites.

the fuel consumption in the diesel generator. Therefore, Bajaur, Rajan Pur, and Karachi show higher carbon emissions, that is 64.71 (tonnes/year). After integrating the renewable resources, the optimal solution for all selected BTS sites is identified. The proposed BTS sites depend purely on renewable generation (solar and wind). Thus, dependency on diesel fuel consumption in the diesel generator reduces and causes less carbon emission. Therefore, in the proposed BTS sites, the minimum carbon emission in the Islamabad BTS site is 0.16 (tonnes/year).

In contrast, the maximum carbon emission noted in the Gilgit BTS site is 5.82 (tonnes/year). Therefore, it can be concluded that these proposed BTS are environmentally friendly with the optimal better performance in technical with the economic perspective.

5.4. *Sensitivity analyses*

Sensitivity analysis facilitates assessing how an ideal system would

behave in the face of various uncertainties, including changes in load, wind speed, inflation rate, discount rate, and solar radiation. This research assesses how these sensitivity variables affect the HES of the BTS Karachi-I (PV-W-B-DG).

5.4.1. Sensitivity to inflation rate (IR) and discount rate (DR)

In this study, Sensitivity analysis was carried out using 9.75 % and 8.5 % as the reference values, respectively, for five altered discount and inflation rates with an estimated variance of ±1.0 %. it can be noted from Fig. 12, that due to the increase in the inflation rate from 6.5 to 10.5, the Levelized cost of energy (LCOE) reduces from 0.123 $/kWh to 0.0962 $/kWh and Net present cost (NPC) increases from 0.1606 $M to 0.19829 $M. The rise in government subsidies given to power producers due to the country's rising level of inflation is the cause of this sharp decline in LCOE. On the other hand, Fig. 12 shows that due to the increase in the discount rate from 7.75.

5.4.2. Sensitivity to wind speed and solar irradiance

Adjusting the average solar radiation and wind speed with an estimated range of ±1.0 % and using the GHI of 5.45 kWh/m^2/day and the wind speed of 5.96 m/s as reference values, respectively, sensitivity analysis was carried out for five different values. The sensitivity analysis by considering LCOE and NPC is shown in Fig. 12 (middle). The sensitivity analysis result illustrates that LCOE and NPC showed an inverse relation with the increase of solar radiation and wind speed. By increasing wind speed from 4.96 (m/s) to 6.96 (m/s), there is a reduction in NPC and LCOE from 0.19177 $M to 0.16417 $M and 0.118 $/kWh to 0.101 $/kWh respectively. On the other side, by increasing solar irradiance from 4.45 (kWh/m^2/day) to 6.45 (kWh/m^2/day), LCOE and NPC reduce from 0.116 $/kWh to 0.104 $/kWh and 0.18927 $M to 0.16902 $M. it can be concluded that NPC and LCOE are much more sensitive to wind speed and solar irradiance.

5.4.3. Sensitivity to load variation

In the future telecom sector, the electrical load can be increased due to an increase in the number of users. Therefore, the sensitivity is analyzed by varying the electrical demand with an expected variation of ±1.0 % by taking the 206.4 KW load as a reference. The graphical presentation is shown in Fig. 12 (bottom). It can be noted that NPC and LCOE showed a direct and inverse relationship with load variation respectively. By increasing the load from 106.4 kWh/day to 306.4 kWh/day, the NPC increased from 0.09944$M to 0.26423 $M, and the LCOE reduced from 0.118 $/kWh to 0.109 $/kWh. This is because the price per unit of any given item rises as the supply decreases. However, the per-unit costs have typically stayed low with a larger amount of the same thing. To guarantee a reliable and sustainable energy surplus with indigenous development, the findings of this research are presented in terms of suggested strategies and policies surrounding NEM. Promoting equality and employment creation can also improve the region's social and environmental characteristics. A hybrid energy system will assure energy security and reliability, especially when it has a variety of various heterogeneous energy supplies. The results of this proposed research produce a policy roadmap and could be a critical factor in helping the region meet its goals for a sustainable energy mix, which have been set by governing bodies as well as private organizations in the power and energy sector.

6. Policy recommendation and future work implication

Pakistan is naturally having a healthy potential for renewable energy resources. According to this research, Pakistan's central and northern zone has a higher dependency on solar radiation. Therefore, all BTS sites in the central and north regions showed strong potential for solar energy with the DG as a backup energy. While the southern zone of Pakistan has more wind speed, some southern sites show dependency on wind energy. Thus, depending upon the area, solar and wind with the battery is the best optimal solution for the telecom sector to shift their load totally onto renewable resource for sustainability. Unfortunately, in Pakistan's telecom sector, all BTS sites depend on the DG, which is harmful to the environment and not economical. In these circumstances, the proposed research could serve as a policy directive to encourage funding for renewable energy initiatives for the telecom sector.

In future work, this study can be expanded with an increase in the number of multiple renewable resources like biomass and hydropower energy. It can be connected with the national grid for net metering. Also, electric vehicles can be integrated with the telecom sector to meet the electrical demand with sustainability and reliability.

7. Conclusions

Diesel generators or traditional grid power supplies run Base Transceiver Stations (BTS) exclusively. Due to the high fuel cost on the global market, Pakistan's BTS of mobile networks, which require a constant supply of electricity, frequently experience power outages, load shedding, high energy costs, and high diesel prices. To solve these issues, many energy production techniques are being researched. This study offered a roadmap for making decisions on the techno-economic viability and sustainability assessment of hybrid systems with a focus on renewable energy for 34 independent, standalone BTS sites.

These 34 BTS sites are taken from all over Pakistan and divided into three regions North, Central, and South. Optimization results in this study concluded that in the north and central regions of Pakistan, solar energy is more dominant than wind energy. Therefore, the optimal configuration for the BTS sites in the north and central region is PV-B-DG. Thus, Mingora BTS-15 showed the least LCOE of 0.1258 $/kWh in the north region with the NPC of $0.080987 million and having operation cost is $2043 with 31 % excess energy. On the other side, the Gilgit BTS-13 has a higher LCOE of 0.2325 $/kWh with the NPC of $0.101247 million and having operation cost of $3511 with 44.6 % excess energy. The only reason is that Gilgit has a lower temperature which reduces the solar peak radiation hours that directly affect the output power of the PV module. In the central region, Lahore BTS-17 and DG Khan BTS-22 showed the least and maximum LCOE of 0.1272 $/kWh and 0.1368 $/kWh respectively. The excess energy in these two sites is 31.2 % and 32.9 % respectively. In the southern region of Pakistan, there is high wind speed. Therefore, five BTS sites showed PV-W-B-DG optimal configuration. In these BTS sites, Karachi-I BTS-24 showed the least LCOE of 0.1096 $/kWh with the NPC of $0.178392 million and also has 31.3 % excess energy. The Rajan Pur BTS-30 is noted as having a higher LCOE of 0.1294$/kWh across this configuration with NPC of $0.210964 million and with 30.9 % excess energy. A comparison of dispatch strategies showed that the LF strategy performed better than the CC strategy in terms of excess energy, operating costs, and payback time and had a 39.55 % lower LCOE. The proposed BTS sites have lower LCOE compared to existing BTS sites, which indicates the proposed system's excellent performance. A system with non-renewable components produces more carbon dioxide than a proposed system that is unfriendly to the environment. In contrast, environmental studies revealed that using PV and wind together provides a more sustainable option with no emissions due to a reduction in fuel consumption. Finally, sensitivity analysis revealed that the LCOE has an inverse correlation with the inflation rate and a direct correlation with the discount rate. Like NPC, LCOE directly correlates with wind speed and solar radiation. In addition, it can be noted that NPC and LCOE showed direct and inverse relationships with an increase in load profile demand. Hybrid BTS sites are, therefore, more economical and environmentally friendly regarding worries about global warming and long-term system functioning with no pollution. In conclusion, building improved BTS sites has positive technical, environmental, and financial effects on Pakistan's telecom industry.

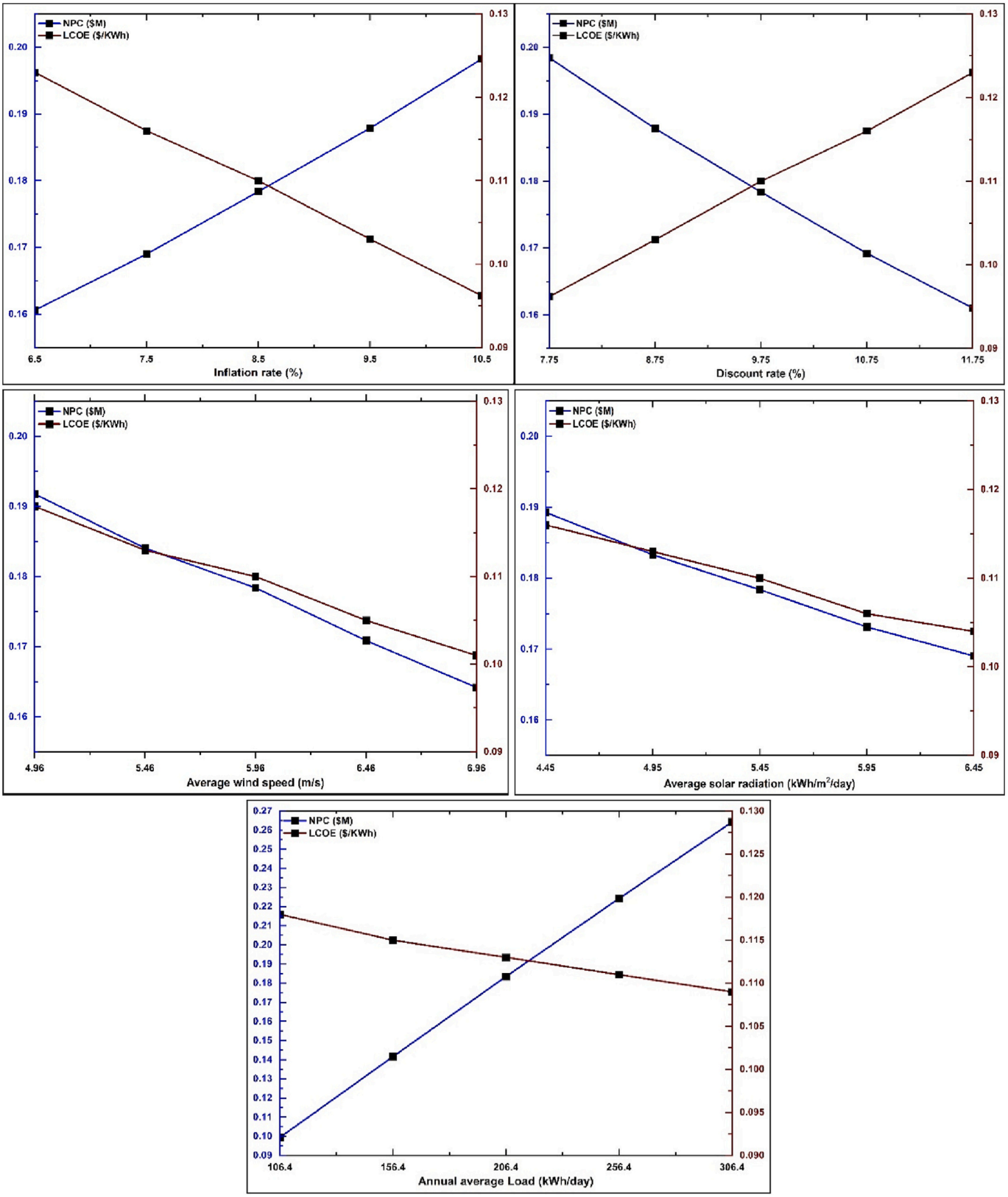


Fig. 12. Sensitivity analysis results on different uncertainty parameters i.e., inflation rate, discount rate, wind speed, solar irradiance, and load demand to 11.75, LCOE increased from 0.0962 $/kWh to 0.123 $/kWh and an overall decrease in NPC from 0.19844 $M to 0.16105 $M.

CRediT authorship contribution statemen

Muhammad Bilal Ali: Conceptualization, Investigation, Methodology, Software, Validation, Visualization, Roles/Writing – original draft, Writing & editing, Formatting.

Syed Ali Abbas Kazmi: Data curation, Formal analysis, Resources, Review & editing, Supervision.

Shahid Nawaz Khan: Conceptualization, Investigation, Methodology, Software.

Muhammad Farasat Abbas: Methodology, Roles/Writing – original draft, Writing & editing, Formatting.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

Table A1
Complete optimization results of proposed and existing under-study BTS sites.

BTS site no.	Project area	Existing infrastructure						Proposed infrastructure							
		Confg.	DG (kW)	BA (no.)	Con. (kW)	DGH (no.)	DGFC (L/yr.)	Confg.	DG (kW)	PV (kW)	WT (no.)	BA (no.)	Con. (kW)	DGH (no.)	DGFC (L/yr.)
Northern zone															
BTS-01	Chakwal	DG-B	3.1	11	0.104	8760	7304	PV-B-DG	3.1	24.8	0	66	3.16	518	349
BTS-02	Islamabad	DG-B	8	11	0.229	8760	18,799	PV-B-DG	8	59.3	0	165	7.98	529	907
BTS-03	Jehlum	DG-B	3.5	11	0.104	8760	8120	PV-B-DG	3.5	25.7	0	77	3.43	480	359
BTS-04	Rawalpindi	DG-B	6.9	11	0.208	8760	16,193	PV-B-DG	6.9	52.1	0	143	6.83	509	752
BTS-05	Talagang	DG-B	2.6	11	0.083	8760	6026	PV-B-DG	2.6	20.4	0	55	2.55	499	279
BTS-06	Taxila	DG-B	6.6	11	0.208	8760	15,633	PV-B-DG	6.6	50.2	0	143	6.64	507	726
BTS-07	Bajaur	DG-B	9.5	11	0.271	8760	22,427	PV-B-DG	9.5	71.7	0	198	9.56	510	1033
BTS-08	Dir	DG-B	4.4	11	0.146	8760	10,421	PV-B-DG	4.4	36.7	0	88	0.182	2056	1956
BTS-09	Kohat	DG-B	4	11	0.125	8760	9398	PV-B-DG	4	31	0	88	4.02	548	469
BTS-10	Nowshera	DG-B	4	11	0.125	8760	9398	PV-B-DG	4	31.2	0	88	3.98	462	402
BTS-11	Peshawar	DG-B	4	11	0.125	8760	9398	PV-B-DG	4	30.8	0	88	3.96	483	417
BTS-12	Abbottabad	DG-B	4.1	11	0.125	8760	9654	PV-B-DG	4.1	32.3	0	88	0.125	597	530
BTS-13	Gilgit	DG-B	2.6	11	0.083	8760	6026	PV-B-DG	2.6	21.3	0	47	2.59	3687	2017
BTS-14	Mingora	DG-B	3.8	11	0.040	8760	8887	PV-B-DG	3.8	27.1	0	77	3.7	545	436
BTS-15	Mirpur	DG-B	2.2	11	0.016	8760	5210	PV-B-DG	2.2	17.4	0	44	2.2	570	268
BTS-16	Muzaffarabad	DG-B	5.9	11	0.053	8760	13,843	PV-B-DG	5.9	44.8	0	121	5.84	663	840
Central zone															
BTS-17	Lahore	DG-B	6.4	11	0.188	8760	15,121	PV-B-DG	6.4	48.9	0	132	6.41	511	708
BTS-18	Sheikhupura	DG-B	4.6	11	0.146	8760	10,272	PV-B-DG	4.6	35.6	0	99	4.63	609	606
BTS-19	Bhakkar	DG-B	3.7	11	0.104	8760	8631	PV-B-DG	3.7	29	0	77	3.68	652	528
BTS-20	Khushab	DG-B	2.2	11	0.083	8760	5210	PV-B-DG	2.2	17.7	0	44	2.21	637	301
BTS-21	Mianwali	DG-B	3.1	11	0.104	8760	7304	PV-B-DG	3.1	24.7	0	66	3.09	551	373
BTS-22	DG Khan	DG-B	2.2	11	0.083	8760	5210	PV-B-DG	2.2	18.6	0	44	2.21	672	323
BTS-23	Layyah	DG-B	2.2	11	0.083	8760	5210	PV-B-DG	2.2	18.5	0	44	2.21	537	303
Southern zone															
BTS-24	Karachi-I	DG-B	9.5	11	0.271	8760	22,427	PV-W-B-DG	9.5	54.2	1	154	9.5	473	775
BTS-25	Karachi-II	DG-B	4.1	11	0.125	8760	9654	PV-W-B-DG	4.1	16.2	1	55	4.18	589	407
BTS-26	Badin	DG-B	2.9	11	0.083	8760	6793	PV-W-B-DG	2.9	10.4	1	33	2.94	670	329
BTS-27	Hyderabad	DG-B	3.8	11	0.125	8760	8887	PV-W-B-DG	3.8	15.7	1	55	3.75	522	344
BTS-28	Mirpur Khas	DG-B	2.2	11	0.083	8760	5210	PV-B-DG	2.2	17.6	0	44	2.2	561	269
BTS-29	Ghotki	DG-B	3.8	11	0.125	8760	8887	PV-B-DG	3.8	30.1	0	77	3.76	630	513
BTS-30	Rajan Pur	DG-B	9.5	11	0.271	8760	22,427	PV-W-B-DG	9.5	66.8	1	176	9.61	645	1353
BTS-31	Rahim Yar Khan	DG-B	3.2	11	0.104	8760	7560	PV-B-DG	3.2	25.9	0	66	3.19	621	433
BTS-32	Sukkur	DG-B	2.2	11	0.083	8760	5210	PV-B-DG	2.2	17	0	44	2.23	536	257
BTS-33	Gawadar	DG-B	2.5	11	0.083	8760	5770	PV-B-DG	2.5	20	0	55	2.43	522	277
BTS-34	Quetta	DG-B	3.5	11	0.104	8760	8120	PV-B-DG	3.5	23.9	0	77	3.42	425	313

Note: PV=Photovoltaic, WT = Wind turbine, DG = Diesel Generator, B = Battery, BA = Battery, Confg. = Configuration, DGH = DG hours, DGFC = DG fuel consumption, Con. = Converter.

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